YEAR 10 MATHEMATICS

Year 10 Trigonometry Part 2

Lesson and Homework Pack

Revision	Right-angled triangles, bearings, and the sine, cosine and area rules	Year 11 Advanced · 5A, 5B, 5I–5K
Lesson T1	Angles of any size	Past papers: 17 marks · 25 min
Lesson T2	Exact values and trigonometric equations	Past papers: 19 marks · 30 min
Lesson T3	The ambiguous case	Past papers: 21 marks · 30 min
Review	Chapter review: Lessons T1 to T3	Past papers: 14 marks · 20 min

This pack follows Topic 11 (Trigonometry C) and covers Trigonometry D (MA5-TRG-P-02). All angles are in degrees. Each lesson runs in the same order: **concept** → **tutor demo** → **your attempt** → **worked example** to keep for homework. The homework comes in four bands: **Easy**, **Medium** and **Hard** questions, then a **past-paper mini exam** to do under exam conditions. Answers and full past-paper solutions are in the separate Homework Answers booklet.

REVISION

Right-angled triangles, bearings, and the sine, cosine and area rules

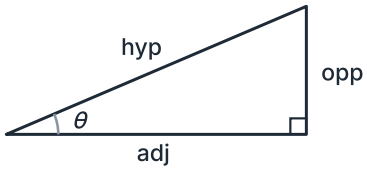
A short review of Topic 11 (Trigonometry C), using the Year 11 Advanced book: right-angled triangles (5A), angles of elevation and bearings (5B), and the sine rule, the area formula and the cosine rule (5I–5K). Every lesson in this pack uses these.

SUMMARY · keep these pages open while you work

THE TRIGONOMETRIC FUNCTIONS FOR ACUTE ANGLES (YEAR 11 ADVANCED, 5A)

Let θ be any acute angle, $0° < \theta < 90°$. Construct a right-angled triangle with an acute angle θ, and label the sides:

- hyp — the *hypotenuse*, the side opposite the right angle,
- opp — the side *opposite* the angle θ,
- adj — the third side, *adjacent* to θ but is not the hypotenuse.



1 The trigonometric functions for an acute angle θ

$$\sin\theta = \frac{\text{opp}}{\text{hyp}} \qquad \cos\theta = \frac{\text{adj}}{\text{hyp}} \qquad \tan\theta = \frac{\text{opp}}{\text{adj}}$$

4 To find an unknown side of a right-angled triangle

1. Start by writing $\dfrac{\text{unknown side}}{\text{known side}} = \ldots$ (place the unknown at top left).
2. Complete the RHS with sin, cos or tan, or the reciprocal of one of these.

5 Finding an unknown angle, given two sides of a right-angled triangle

Work out from the known sides which one of $\cos\theta$, $\sin\theta$ or $\tan\theta$ to use.

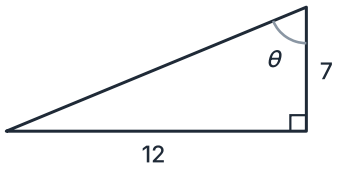
LEFT OUT OF THIS SUMMARY Box 1 in the book also defines cosec, sec and cot, the reciprocals of sin, cos and tan. They are not part of the Year 10 course, so this pack does not use them.

WORKED EXAMPLE **Year 11 Advanced Example 2 (5A)**

Find θ in the triangle drawn to the right.

SOLUTION

The given sides are the opposite and the adjacent sides, so $\tan\theta$ is known.



$$\tan\theta = \frac{12}{7} \qquad \left(\frac{\text{opposite}}{\text{adjacent}}\right)$$

$$\theta \doteqdot 59°45'.$$

STEPS THE BOOK SKIPS: EXAMPLE 2 On the calculator, $\tan^{-1}\left(\frac{12}{7}\right) = 59.743\ldots°$. The decimal part is $0.743\ldots \times 60' = 44.6'$, which rounds to $45'$, so $\theta \doteqdot 59°45'$. The sign $\doteqdot$ means "approximately equal to".

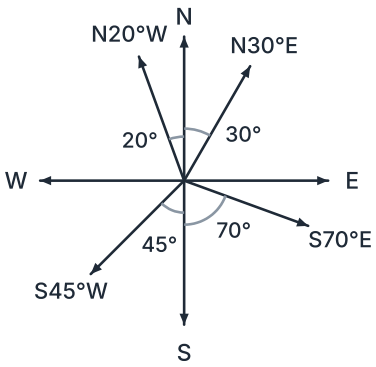
ANGLES OF ELEVATION AND DEPRESSION, AND BEARINGS (YEAR 11 ADVANCED, 5B)

When the figure has two or more triangles, always name the triangle that you are working in.

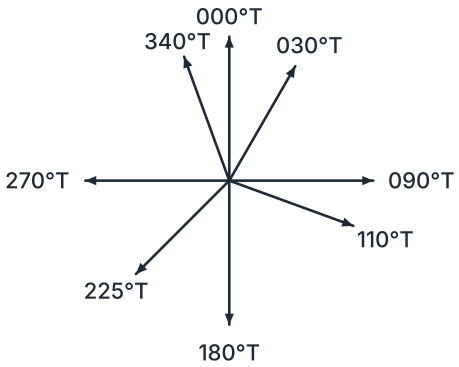
Angles of elevation and depression. Angles of elevation and depression are always measured from the horizontal, and are always acute angles.

Compass bearings and true bearings. *Compass bearings* are based on north, south, east and west. Any other direction is specified by the deviation from north or south towards the east or west. The diagram to the left below gives four examples. Note that S45°W can also be written simply as SW (that is, south west).

True bearings are measured clockwise from north (not anticlockwise as in the coordinate plane). The diagram on the right below gives the same four directions expressed as true bearings. Three digits are used, even for angles less than 100°.



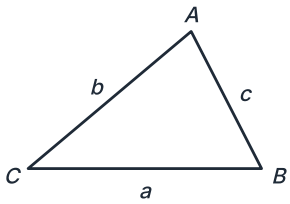
Compass bearing



True bearing

THE SINE RULE AND THE AREA FORMULA (YEAR 11 ADVANCED, 5I)

The usual statement of all three rules uses the convention shown in the diagram to the right. The vertices are named with upper-case letters, then each side takes its name from the lower-case letter of the opposite vertex.



30 The sine rule. In any triangle ABC,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

'The ratio of each side to the sine of the opposite angle is constant.'

31 Using the sine rule to find a side. In the AAS congruence situation:

$$\frac{\text{unknown side}}{\text{sine of its opposite angle}} = \frac{\text{known side}}{\text{sine of its opposite angle}}.$$

Always place the unknown side at the top left of the equation.

If two angles of a triangle are known, so is the third, because the angles add to 180°.

32 The area formula. In any triangle ABC,

$$\text{area } \Delta ABC = \tfrac{1}{2}bc\sin A.$$

'The area of a triangle is half the product of any two sides times the sine of the included angle'.

THE COSINE RULE (YEAR 11 ADVANCED, 5J)

35 The cosine rule. In any triangle ABC,

$$a^2 = b^2 + c^2 - 2bc\cos A.$$

'The square of any side of a triangle equals: the sum of the squares of the other two sides, minus twice the product of those sides and the cosine of their included angle.'

37 Using the cosine rule to find a side. In the SAS congruence situation: (square of any side) = (sum of squares of other two sides) − (twice the product of those sides) × (cosine of their included angle).

38 Using the cosine rule to find an angle. In the SSS congruence situation: Substitute into the cosine rule and solve for $\cos\theta$.

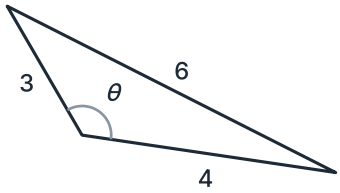
39 The cosine rule with $\cos A$ as subject. In any triangle ABC,

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}.$$

Notice that $\cos\theta$ is positive when θ is acute, and is negative when θ is obtuse. Hence, there is only ever one solution for the unknown angle, unlike the situation for the sine rule, in which there are often two possible angles.

WORKED EXAMPLE **Year 11 Advanced Example 29 (5J)**

Find θ in the triangle drawn to the right.



SOLUTION

Substituting into the cosine rule,

$$6^2 = 3^2 + 4^2 - 2 \times 3 \times 4 \times \cos\theta$$
$$24\cos\theta = -11$$
$$\cos\theta = -\frac{11}{24}$$
$$\theta \doteqdot 117°17'.$$

OR

Using the box 39 formula,

$$\cos\theta = \frac{3^2 + 4^2 - 6^2}{2 \times 3 \times 4}$$
$$\cos\theta = \frac{-11}{24}$$
$$\theta \doteqdot 117°17'.$$

STEPS THE BOOK SKIPS: EXAMPLE 29 On the left, $36 = 9 + 16 - 24\cos\theta = 25 - 24\cos\theta$, so $24\cos\theta = 25 - 36 = -11$. On the right, $9 + 16 - 36 = -11$. The cosine is negative, so θ is obtuse: $\cos^{-1}\left(-\frac{11}{24}\right) = 117.28\ldots° \doteqdot 117°17'$.

WHICH RULE TO USE (YEAR 11 ADVANCED, 5K)

40 The sine, cosine and area rules and the standard congruence tests. In a right-angled triangle, use simple trigonometry and Pythagoras. Otherwise:

- AAS: Use the sine rule to find each of the other two sides.
- ASS: [The ambiguous situation] Use the sine rule to find the unknown angle opposite a known side. There may or may not be two possible solutions.
- SAS: Use the cosine rule to find the third side. Use the area formula to find the area.
- SSS: Use the cosine rule to find any angle.

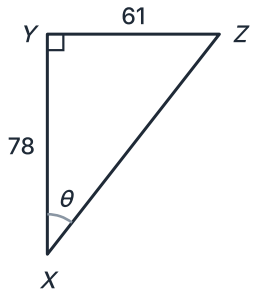
WHERE THIS PACK GOES NEXT Lesson T1 extends sin, cos and tan to angles of any size, including obtuse angles. Lesson T3 deals with the ambiguous ASS situation.

WARM-UP · answers are in the answers booklet

Each question is from the Year 11 Advanced exercise named, with some parts left out. A calculator is needed.

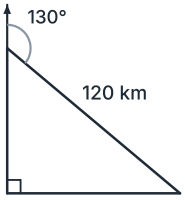
1 Year 11 Advanced Exercise 5B Q4

A ship sails 78 nautical miles due north from X to Y, then 61 nautical miles due east from Y to Z. Find θ, the bearing of Z from X, correct to the nearest degree.



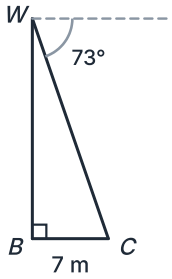
2 Year 11 Advanced Exercise 5B Q7

Eleni drives 120 km on a bearing of 130°T. She then drives due west until she is due south of her starting point. How far is she from her starting point, correct to the nearest kilometre?



3 Year 11 Advanced Exercise 5B Q8

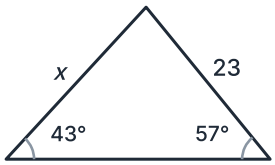
John is looking out of the window W at a car C parked on the street below. If the angle of depression of C from W is 73° and the car is 7 metres from the base B of the building, find the height WB of the window, correct to the nearest metre.



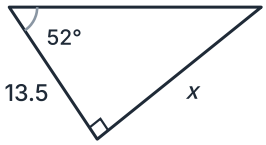
4 Year 11 Advanced Exercise 5K Q1

Use right-angled triangle trigonometry, the sine rule or the cosine rule to find x in each triangle, correct to one decimal place.

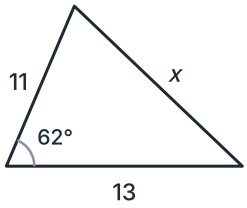
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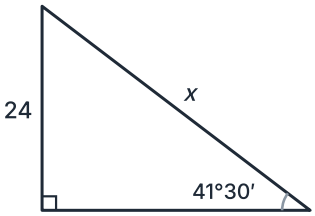
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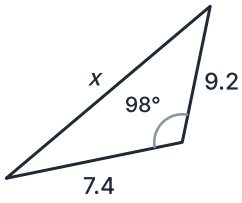
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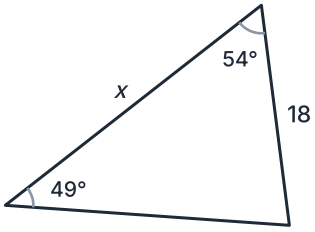
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e



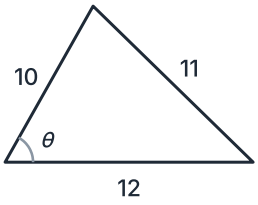
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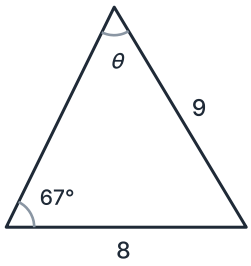
5 Year 11 Advanced Exercise 5K Q2

Use right-angled triangle trigonometry, the sine rule or the cosine rule to find θ in each triangle, correct to the nearest degree.

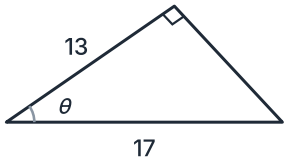
a



b



c

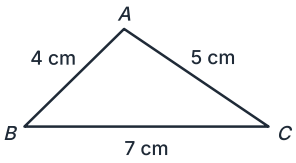


6 Year 11 Advanced Exercise 5K Q7

In ΔABC, $AB = 4$ cm, $BC = 7$ cm and $CA = 5$ cm.

a Use the cosine rule to find $\angle ABC$, correct to the nearest minute.

b Hence calculate the area of ΔABC, correct to the nearest square centimetre.



LESSON T1

Angles of any size

The definitions of sin, cos and tan are extended from acute angles to angles of any size, positive or negative, using a circle in the coordinate plane. The quadrant of the angle and the ASTC rule give the sign, and the related acute angle gives the size, so every value comes back to an acute angle. The lesson ends with the graphs of sin, cos and tan, and the angle of inclination of a line, where the gradient is $\tan\alpha$. Sources: Year 11 Advanced book 5D, 5E and 6B; Maths Pathway sheets *The Unit Circle*, *Using Symmetry on the Unit Circle* and *Graphs of Trigonometric Functions*.

CONCEPT 1 · angles of any size and their rays

TRIGONOMETRIC FUNCTIONS OF A GENERAL ANGLE (YEAR 11 ADVANCED, 5D)

The definitions of the trigonometric functions given in Section 5A only apply to acute angles, because in a right-angled triangle, both other angles are acute angles.

This section introduces more general definitions based on circles in the coordinate plane (whose equations are Pythagoras' theorem, as we saw in Section 3G). The new definitions will apply to any angle, but will, of course, give the same values as the previous definitions for acute angles.

Putting a general angle on the coordinate plane. Let θ be any angle — possibly negative, possibly obtuse or reflex, possibly greater than 360°. We shall associate with θ a ray with vertex at the origin.

8 THE RAY CORRESPONDING TO θ

- The positive direction of the x-axis is the ray representing the angle 0°.
- For all other angles, rotate this ray anticlockwise through an angle θ.

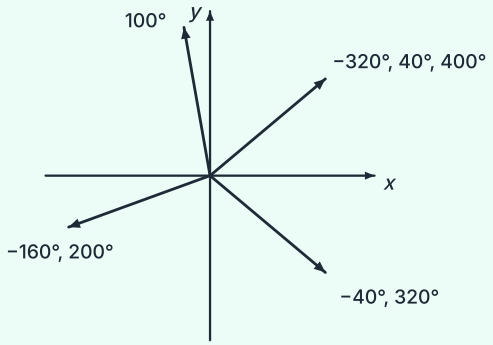
If the angle is negative, the ray is rotated backwards, which means clockwise.

Here are some examples of angles and the corresponding rays. The angles have been written at the ends of the arrows representing the rays.

Notice that one ray can correspond to many angles. For example, all the following angles have the same ray as 40°:

..., −680°, −320°, 40°, 400°, 760°, ...

A given ray thus corresponds to infinitely many angles, all differing by multiples of 360°. The relation from rays to angles is thus one-to-many.



9 CORRESPONDING ANGLES AND RAYS

- To each angle, there corresponds exactly one ray.
- To each ray, there correspond infinitely many angles, all differing from each other by multiples of 360°.

FINDING ANGLES WITH THE SAME RAY

Add or subtract 360° as many times as you need. One full turn brings the ray back to where it started. For example, $400° - 360° = 40°$ and $40° - 360° = -320°$. To find the angle between 0° and 360° with the same ray as a negative angle, keep adding 360° until the answer is positive.

TUTOR DEMO **Angles with the same ray**

a Write down two positive angles and two negative angles that have the same ray as 50°.

b Which angle between 0° and 360° has the same ray as: **i** −120°, **ii** 470°?

YOUR TURN **Same method**

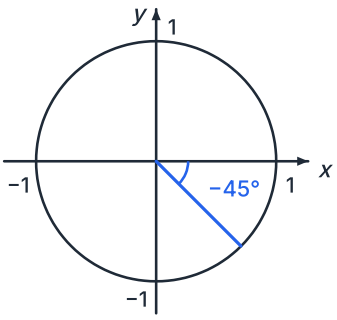
a Write down two positive angles and two negative angles that have the same ray as 70°.

b Which angle between 0° and 360° has the same ray as: **i** −150°, **ii** 520°?

WORKED EXAMPLE **From the text: Maths Pathway, *Graphs of Trigonometric Functions***

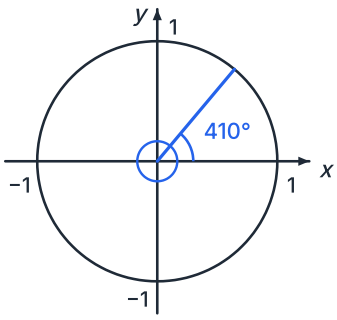
Equivalent angles on the unit circle. The unit circle can measure negative angles and angles greater than 360°. If two different angles look the same on the unit circle, we can say that they are equivalent angles.

Negative angles are measured in the clockwise direction.



−45° is equivalent to 315°

Angles greater than 360° continue in the anti-clockwise direction.



410° is equivalent to 50°

STEPS THE SOURCE SKIPS: EQUIVALENT ANGLES

$-45° + 360° = 315°$, so −45° is equivalent to 315°. $410° - 360° = 50°$, so 410° is equivalent to 50°. "Equivalent angles" is the Maths Pathway name for angles with the same ray.

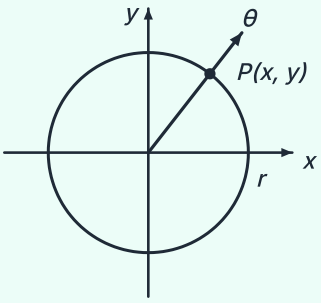
CONCEPT 2 · sin, cos and tan of any angle

DEFINING THE TRIGONOMETRIC FUNCTIONS FOR GENERAL ANGLES (YEAR 11 ADVANCED, 5D)

Let θ be any angle, positive or negative.

Construct a circle with centre the origin and any positive radius r. Let the ray corresponding to θ intersect the circle at the point $P(x, y)$.

The six trigonometric functions are now defined in terms of x, y and r as follows:



10 DEFINITIONS OF THE TRIGONOMETRIC FUNCTIONS

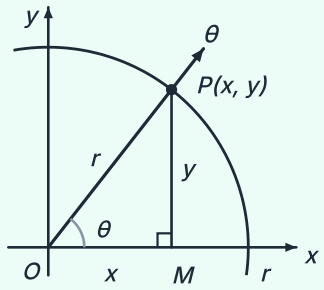
$$\sin\theta = \frac{y}{r} \qquad \cos\theta = \frac{x}{r} \qquad \tan\theta = \frac{y}{x}$$

Note: We chose r to be 'any positive radius'. If a different radius were chosen, the two figures would be similar, so the lengths x, y and r would stay in the same ratio. Because the definitions depend only on the ratios of the lengths, the values of the trigonometric functions would not change.

In particular, we may use a circle of radius 1 in the definitions. When this is done, however, we would then lose the intuition that a trigonometric function is not a length, but is the ratio of two lengths.

Agreement with the earlier definition. Let θ be an acute angle. Construct the ray corresponding to θ. Let the perpendicular from P meet the x-axis at M. Then $\theta = \angle POM$, so relating the sides to the angle θ,

hypotenuse = $OP = r$, opposite = $PM = y$, adjacent = $OM = x$.



Hence the old and the new definitions are in agreement.

Note: Most people find that the diagram above is the easiest way to learn the new definitions of the trigonometric functions. Take the old definitions in terms of hypotenuse, opposite and adjacent sides, and make the replacements,

hypotenuse $\longleftrightarrow r$, opposite $\longleftrightarrow y$, adjacent $\longleftrightarrow x$.

LEFT OUT OF THIS LESSON: COSEC, SEC AND COT

The book defines six functions here. The other three, cosec, sec and cot, are the reciprocals of sin, cos and tan. They are not in the Year 10 course, so this pack leaves them out of Boxes 10, 11, 12 and 14 and of the graphs. Where a sentence of the book mentions them, read only the part about sin, cos and tan.

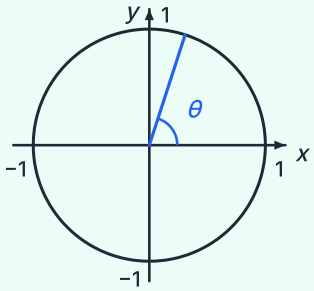
THE UNIT CIRCLE (MATHS PATHWAY, *THE UNIT CIRCLE* AND *USING SYMMETRY ON THE UNIT CIRCLE*)

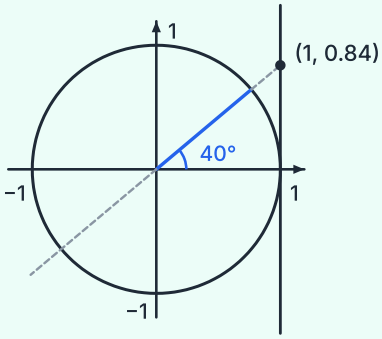
Note: the values in this module have been rounded to two decimal places.

The unit circle has a radius of 1. Angles are measured starting from the positive side of the x-axis.

The unit circle has a tangent that touches at the point $(1, 0)$. The tangent of any angle $(\tan\theta)$ is the y-value where the angle meets the tangent to the circle.

$\tan 40° = 0.84$





WHY THE UNIT CIRCLE IS USEFUL

With $r = 1$, Box 10 becomes $\sin\theta = \frac{y}{1} = y$ and $\cos\theta = \frac{x}{1} = x$. So the point where the ray meets the unit circle is $(\cos\theta, \sin\theta)$: cos is the x-coordinate and sin is the y-coordinate. Dividing, $\tan\theta = \frac{y}{x} = \frac{\sin\theta}{\cos\theta}$. On the tangent line $x = 1$, the ray reaches the height $\tan\theta$, because the triangles are similar: $\frac{\text{height}}{1} = \frac{y}{x}$.

TUTOR DEMO **Values from a point on the ray**

The ray of the angle θ meets the circle with centre O and radius 13 at the point $P(-5, 12)$.

a Check that P lies on the circle, that is, $OP = 13$.

b Write down the values of $\sin\theta$, $\cos\theta$ and $\tan\theta$.

YOUR TURN **Same method**

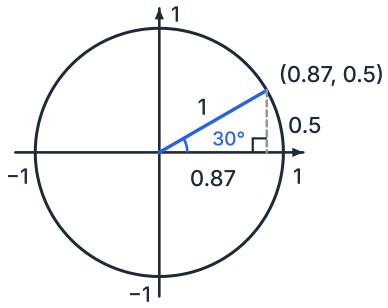
The ray of the angle θ meets the circle with centre O and radius 17 at the point $P(8, -15)$.

a Check that P lies on the circle, that is, $OP = 17$.

b Write down the values of $\sin\theta$, $\cos\theta$ and $\tan\theta$.

WORKED EXAMPLE **From the text: Maths Pathway, *The Unit Circle* Q4 and Q5**

4. An angle of 30° touches the unit circle at the point $(0.87, 0.5)$. Drawing a line from the point straight down to the x-axis creates a right triangle.



a) Copy the diagram and label the lengths of each triangle side.

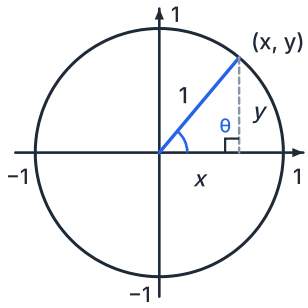
b) Use the diagram to find: $\sin(30°)$, $\cos(30°)$.

SOLUTION

a) See the diagram.

b) $\sin(30°) = \dfrac{0.5}{1} = 0.5$ $\qquad$ $\cos(30°) = \dfrac{0.87}{1} = 0.87$

5. The angle θ touches the unit circle at the point (x, y). Drawing a line from the point straight down to the x-axis creates a right triangle.



a) Copy the diagram and label the lengths of each triangle side.

b) Use the diagram to find $\sin\theta$, $\cos\theta$ and $\tan\theta$. Leave your answers in terms of x and y.

c) Use your answers from part b) to write $\tan\theta$ in terms of $\sin\theta$ and $\cos\theta$.

SOLUTION

a) See the diagram.

b) $\sin\theta = \dfrac{y}{1} = y$ $\qquad$ $\cos\theta = \dfrac{x}{1} = x$ $\qquad$ $\tan\theta = \dfrac{y}{x}$

c) $\tan\theta = \dfrac{y}{x}$ $\quad \sin\theta = y$ $\quad \cos\theta = x$ $\quad \tan\theta = \dfrac{\sin\theta}{\cos\theta}$

STEPS THE SOURCE SKIPS: Q4 AND Q5

In Q4 the right triangle has hypotenuse 1 (the radius), adjacent side 0.87 and opposite side 0.5, so $\sin 30° = \dfrac{\text{opp}}{\text{hyp}} = \dfrac{0.5}{1}$ and $\cos 30° = \dfrac{\text{adj}}{\text{hyp}} = \dfrac{0.87}{1}$. In Q5(c), replace y by $\sin\theta$ and x by $\cos\theta$ in $\tan\theta = \dfrac{y}{x}$.

CONCEPT 3 · boundary angles

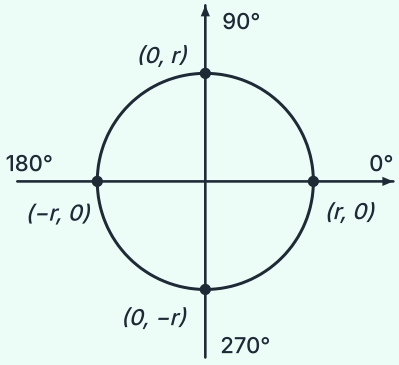
BOUNDARY ANGLES (YEAR 11 ADVANCED, 5D)

Integer multiples of 90°, that is

…, −90°, 0°, 90°, 180°, 270°, 360°, 450°, …

are called *boundary angles* because they lie on the boundaries between quadrants.

The values of the trigonometric functions at these boundary angles are not always defined, and are 0, 1 or −1 when they are defined. The diagram to the right can be used to calculate them, and the results are shown in the table below (where the star * indicates that the value is undefined).



11 THE BOUNDARY ANGLES

θ	0°	90°	180°	270°
x	r	0	$-r$	0
y	0	r	0	$-r$
r	r	r	r	r
$\sin\theta$	0	1	0	−1
$\cos\theta$	1	0	−1	0
$\tan\theta$	0	*	0	*

In practice, the answer to any question about the values of the trigonometric functions at these boundary angles should be read off the graphs of the functions. These graphs need to be known very well indeed.

The domains of the trigonometric functions. The trigonometric functions are defined everywhere except where the denominator is zero.

12 DOMAINS OF THE TRIGONOMETRIC FUNCTIONS

- $\sin\theta$ and $\cos\theta$ are defined for all angles θ.
- $\tan\theta$ and $\sec\theta$ are undefined when $x = 0$, that is, when $\theta = \ldots, -90°, 90°, 270°, 450°, \ldots$

READING THE TABLE

At 90° the point is $(0, r)$, so $\tan 90° = \dfrac{y}{x} = \dfrac{r}{0}$, which is undefined: you cannot divide by zero. A boundary angle outside 0° to 270° has the same values as the angle with the same ray. For example, 450° has the same ray as 90°, and −90° has the same ray as 270°.

TUTOR DEMO **Boundary angles**

Use the boundary-angle diagram to find $\sin\theta$, $\cos\theta$ and $\tan\theta$ (if they exist) for:

a $\theta = 450°$

b $\theta = -90°$

YOUR TURN **Same method**

Use the boundary-angle diagram to find $\sin\theta$, $\cos\theta$ and $\tan\theta$ (if they exist) for:

a $\theta = 540°$

b $\theta = -270°$

WORKED EXAMPLE

From the text: Maths Pathway, *Using Symmetry on the Unit Circle* Q2(c) and (d)

The following questions are related to the tangent of an angle.

c) List two angles on the unit circle where $\tan\theta = 0$.

d) List two angles on the unit circle that will never touch the tangent line.

SOLUTION

c) 0° and 180° touch the tangent to the circle on the x-axis, so $\tan(0°) = 0$ and $\tan(180°) = 0$.

d) 90° and 270° extend along the y-axis, so they will never touch the tangent to the circle.
Note: for these angles we say that $\tan\theta$ is undefined.

STEPS THE SOURCE SKIPS: Q2(C) AND (D)

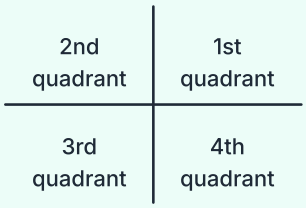
The ray for 180° points left along the x-axis. Extended back through O, it is the x-axis itself, which meets the tangent line $x = 1$ at $(1, 0)$, at height 0. The rays for 90° and 270° lie along the y-axis, which is parallel to the tangent line $x = 1$, so they never meet it. This agrees with Box 11.

CONCEPT 4 · quadrants and the ASTC rule

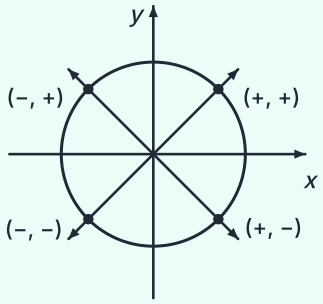
QUADRANT, SIGN, AND RELATED ACUTE ANGLE (YEAR 11 ADVANCED, 5E)

Symmetry is an essential aspect of trigonometric functions. In this section we will use symmetry to express the values of the trigonometric functions of any angle in terms of trigonometric functions of acute angles.

The diagram shows the conventional anticlockwise numbering of the four quadrants of the coordinate plane. Acute angles are in the first quadrant, obtuse angles are in the second, and reflex angles are in the third and fourth.



The signs of the trigonometric functions. The signs of the trigonometric functions depend only on the signs of x and y. (The radius r is always positive.) The signs of x and y depend in turn only on the quadrant in which the ray lies. Thus we can easily compute the signs of the trigonometric functions from the accompanying diagram and the definitions.



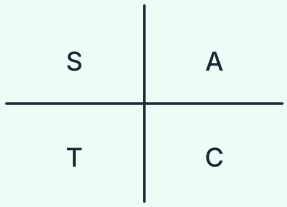
Quadrant	1	2	3	4
x	+	−	−	+
y	+	+	−	−
r	+	+	+	+
$\sin\theta$	+	+	−	−
$\cos\theta$	+	−	−	+
$\tan\theta$	+	−	+	−

In NSW, these results are usually remembered by the phrase:

14 SIGNS OF THE TRIGONOMETRIC FUNCTIONS

'All Stations To Central'

indicating that the four letters A, S, T and C are placed successively in the four quadrants as shown. The significance of the letters is:



A means *all* six functions are positive.
S means only *sine* (and cosecant) are positive.
T means only *tangent* (and cotangent) are positive.
C means only *cosine* (and secant) are positive.

Three graphs of the trigonometric functions were constructed in Exercise 5D, and all six are drawn together at the end of this section. Study each of them to see how the table of signs above, and the ASTC rule, agree with your observations about when the graph is above the x-axis and when it is below.

USING ASTC WITH ANY ANGLE

First find the quadrant of the ray. For a negative angle or an angle greater than 360°, first add or subtract multiples of 360° to get an angle between 0° and 360° with the same ray. Then read the letter: in quadrant 1 all three are positive, in quadrant 2 only sin, in quadrant 3 only tan, in quadrant 4 only cos. The other two are negative.

TUTOR DEMO **Quadrant and signs**

For each angle, state its quadrant, then use the ASTC rule to give the signs of $\sin\theta$, $\cos\theta$ and $\tan\theta$.

a $\theta = 115°$

b $\theta = 250°$

c $\theta = -60°$

YOUR TURN **Same method**

For each angle, state its quadrant, then use the ASTC rule to give the signs of $\sin\theta$, $\cos\theta$ and $\tan\theta$.

a $\theta = 165°$

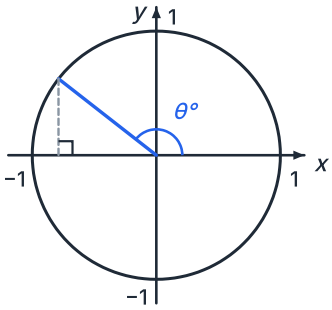
b $\theta = 320°$

c $\theta = 400°$

WORKED EXAMPLE **From the text: Maths Pathway, *The Unit Circle* Q7**

The angle in the following diagram is unknown.
Which of the following is correct?

A) $\sin\theta$ is negative, $\cos\theta$ is positive, and $\tan\theta$ is negative.
B) $\sin\theta$ is positive, $\cos\theta$ is negative, and $\tan\theta$ is positive.
C) $\sin\theta$ is positive, $\cos\theta$ is negative, and $\tan\theta$ is negative.
D) There is not enough information to know if $\sin\theta$, $\cos\theta$ and $\tan\theta$ are positive or negative.



SOLUTION

C) $\sin\theta$ is positive, $\cos\theta$ is negative, and $\tan\theta$ is negative.

STEPS THE SOURCE SKIPS: Q7

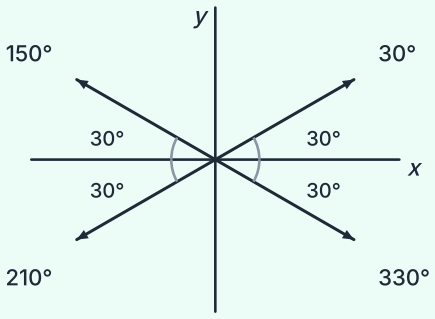
The ray ends in the second quadrant, where $x < 0$ and $y > 0$. So $\sin\theta = y$ is positive, $\cos\theta = x$ is negative, and $\tan\theta = \dfrac{y}{x}$ is a positive divided by a negative, which is negative. This is the S quadrant: only sine is positive.

CONCEPT 5 · the related acute angle

THE QUADRANT AND THE RELATED ACUTE ANGLE (YEAR 11 ADVANCED, 5E)

The diagram to the right shows the four rays corresponding to the four angles 30°, 150°, 210° and 330°.

These four rays lie in each of the four quadrants of the plane, and they all make the same acute angle of 30° with the x-axis. The four rays are thus the reflections of each other in the two axes.



13 QUADRANT AND RELATED ACUTE ANGLE

Let θ be any angle.

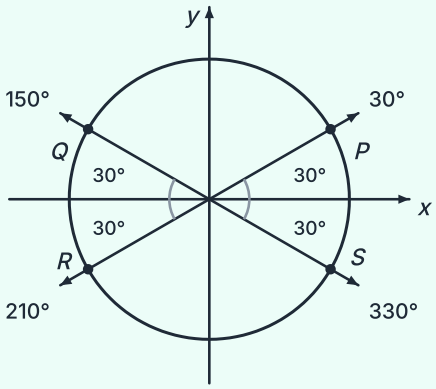
- The *quadrant* of θ is the quadrant (1, 2, 3 or 4) in which the ray lies.
- The *related acute angle* of θ is the acute angle between the ray and the x-axis.

Each of the four angles above has the same related acute angle 30°. Notice that θ and its related angle are only the same when θ is an acute angle.

The angle and the related acute angle. In the diagram to the right, a circle of radius r has been added to the earlier diagram showing the four angles 30°, 150°, 210° and 330°.

The four points P, Q, R and S where the four rays meet the circle are all reflections of each other in the x-axis and y-axis. Because of this symmetry, the coordinates of these four points are identical, apart from their signs.

Hence the trigonometric functions of these angles will all be the same too, except that the signs may be different.



15 THE ANGLE AND THE RELATED ACUTE ANGLE

- The trigonometric functions of any angle θ are the same as the trigonometric functions of its related acute angle, apart from a possible change of sign.
- The sign is best found using the ASTC diagram.

Evaluating the trigonometric functions of any angle. This gives a straightforward way of evaluating the trigonometric functions of any angle.

16 TRIGONOMETRIC FUNCTIONS OF ANY ANGLE

Draw a quadrant diagram then:

- Place the ray in the correct quadrant, and use the ASTC rule to work out the sign of the answer.
- Find the related acute angle, and work out the value of the trigonometric function at this related angle.

REFERENCE ANGLES (MATHS PATHWAY, *USING SYMMETRY ON THE UNIT CIRCLE*)

Every angle on the unit circle has a reference angle in the first quadrant based on the triangle it makes with the x-axis.

The values from the first quadrant will help us figure out the values of $\sin\theta$, $\cos\theta$, and $\tan\theta$ for other angles.

For example, in question 3 you used the angle 40° as a reference for three other angles (140°, 220° and 320°).

FINDING THE RELATED ACUTE ANGLE

"Reference angle" is the Maths Pathway name for the related acute angle. For an angle θ between 0° and 360°:

- quadrant 1: the related angle is θ; quadrant 2: $180° - \theta$;
- quadrant 3: $\theta - 180°$; quadrant 4: $360° - \theta$.

For any other angle, first add or subtract multiples of 360°.

EXACT VALUES YOU NEED NOW (FROM 5A, TAUGHT FULLY IN LESSON T2)

θ	$\sin\theta$	$\cos\theta$	$\tan\theta$
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$

This is the table of exact values in Box 2 of the Year 11 book (sin, cos and tan columns only). Lesson T2 shows where the values come from.

TUTOR DEMO **Exact values of any angle**

Find the exact value of:

a $\sin 480°$

b $\cos(-135°)$

c $\tan 660°$

YOUR TURN **Same method**

Find the exact value of:

a $\sin 600°$

b $\cos(-330°)$

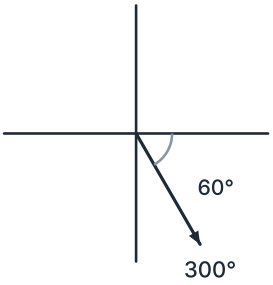
c $\tan 495°$

WORKED EXAMPLE 10 **From the text: keep for homework**

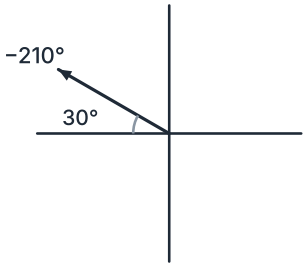
Find the exact value of:

a $\tan 300°$ b $\sin(-210°)$ c $\cos 570°$

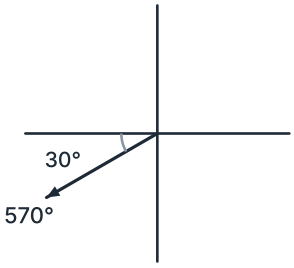
SOLUTION



a 300° is in quadrant 4, the related angle is 60°, so $\tan 300° = -\tan 60°$
$= -\sqrt{3}.$



b −210° is in quadrant 2, the related angle is 30°, so
$\sin(-210°) = +\sin 30°$
$= \frac{1}{2}.$



c 570° is in quadrant 3, the related angle is 30°, so $\cos 570° = -\cos 30°$
$= -\frac{\sqrt{3}}{2}.$

Note: A calculator will give approximate values of the trigonometric functions without any need to find the related acute angle. It will *not* give exact values, however, when these values involve surds.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 10

a 300° is between 270° and 360°, so it is in quadrant 4, and the related angle is $360° - 300° = 60°$. Quadrant 4 is the C quadrant, so tan is negative.

b $-210° + 360° = 150°$, so −210° has the same ray as 150°, in quadrant 2. The related angle is $180° - 150° = 30°$. Quadrant 2 is the S quadrant, so sin is positive.

c $570° - 360° = 210°$, in quadrant 3. The related angle is $210° - 180° = 30°$. Quadrant 3 is the T quadrant, so cos is negative.

CONCEPT 6 · obtuse angles and formulae with general angles

GENERAL ANGLES WITH PRONUMERALS (YEAR 11 ADVANCED, 5E)

This quadrant-diagram method can be used to generate formulae for expressions such as $\sin(180° + A)$ or $\cot(360° - A)$. The trick is to place A on the quadrant diagram *as if it were acute*.

17 SOME FORMULAE WITH GENERAL ANGLES

$$\sin(180° - A) = \sin A \qquad \sin(180° + A) = -\sin A \qquad \sin(360° - A) = -\sin A$$
$$\cos(180° - A) = -\cos A \qquad \cos(180° + A) = -\cos A \qquad \cos(360° - A) = \cos A$$
$$\tan(180° - A) = -\tan A \qquad \tan(180° + A) = \tan A \qquad \tan(360° - A) = -\tan A$$

Some people prefer to learn this list of identities to evaluate trigonometric functions, but this seems unnecessary when the quadrant-diagram method is so clear.

OBTUSE ANGLES IN THE YEAR 10 COURSE

If A is obtuse, then $180° - A$ is its related acute angle, and A is in quadrant 2, where only sin is positive. So, for any obtuse angle A,

$$\sin A = \sin(180° - A)$$
$$\cos A = -\cos(180° - A)$$
$$\tan A = -\tan(180° - A)$$

These are the first column of Box 17 read the other way. For example, $\sin 150° = \sin 30°$ and $\cos 150° = -\cos 30°$.

Negative angles. The ray for $-A$ is the reflection of the ray for A in the x-axis: x stays the same and y changes sign. So $\sin(-A) = -\sin A$, $\cos(-A) = \cos A$ and $\tan(-A) = -\tan A$.

TUTOR DEMO **Using a reference angle**

Given $\sin 35° \fallingdotseq 0.57$, $\cos 35° \fallingdotseq 0.82$ and $\tan 35° \fallingdotseq 0.70$, write down without a calculator:

a $\sin 145°$, $\cos 145°$ and $\tan 145°$

b $\sin 325°$, $\cos 325°$ and $\tan 325°$

YOUR TURN **Same method**

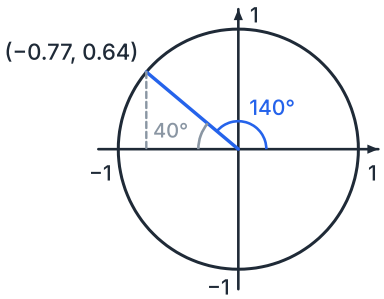
Given $\sin 55° \fallingdotseq 0.82$, $\cos 55° \fallingdotseq 0.57$ and $\tan 55° \fallingdotseq 1.43$, write down without a calculator:

a $\sin 125°$, $\cos 125°$ and $\tan 125°$

b $\sin 235°$, $\cos 235°$ and $\tan 235°$

WORKED EXAMPLE **From the text: Maths Pathway, *Using Symmetry on the Unit Circle* Q3(a)**

The angle 40° has been reflected into other quadrants on the unit circle. $\sin(40°) = 0.64$, $\cos(40°) = 0.77$, $\tan(40°) = \dfrac{\sin(40°)}{\cos(40°)} = 0.84$.



a) i) Copy the diagram and fill in the main angle and the missing coordinates.

ii) Write $\sin\theta$, $\cos\theta$, and $\tan\theta$ for the main angle.

SOLUTION

i) See the diagram: the main angle is 140° and the point is $(-0.77, 0.64)$.

ii) $\sin(140°) = \sin(40°) = 0.64$

$\cos(140°) = -\cos(40°) = -0.77$

$$\tan(140°) = \frac{\sin(40°)}{-\cos(40°)} = -\tan(40°)$$
$$= -0.84$$

STEPS THE SOURCE SKIPS: Q3(A)

The ray for 140° is the reflection of the ray for 40° in the y-axis, because $180° - 40° = 140°$. Reflecting in the y-axis changes the sign of x only, so the point $(0.77, 0.64)$ becomes $(-0.77, 0.64)$. Then $\sin 140°$ is the y-coordinate and $\cos 140°$ is the x-coordinate.

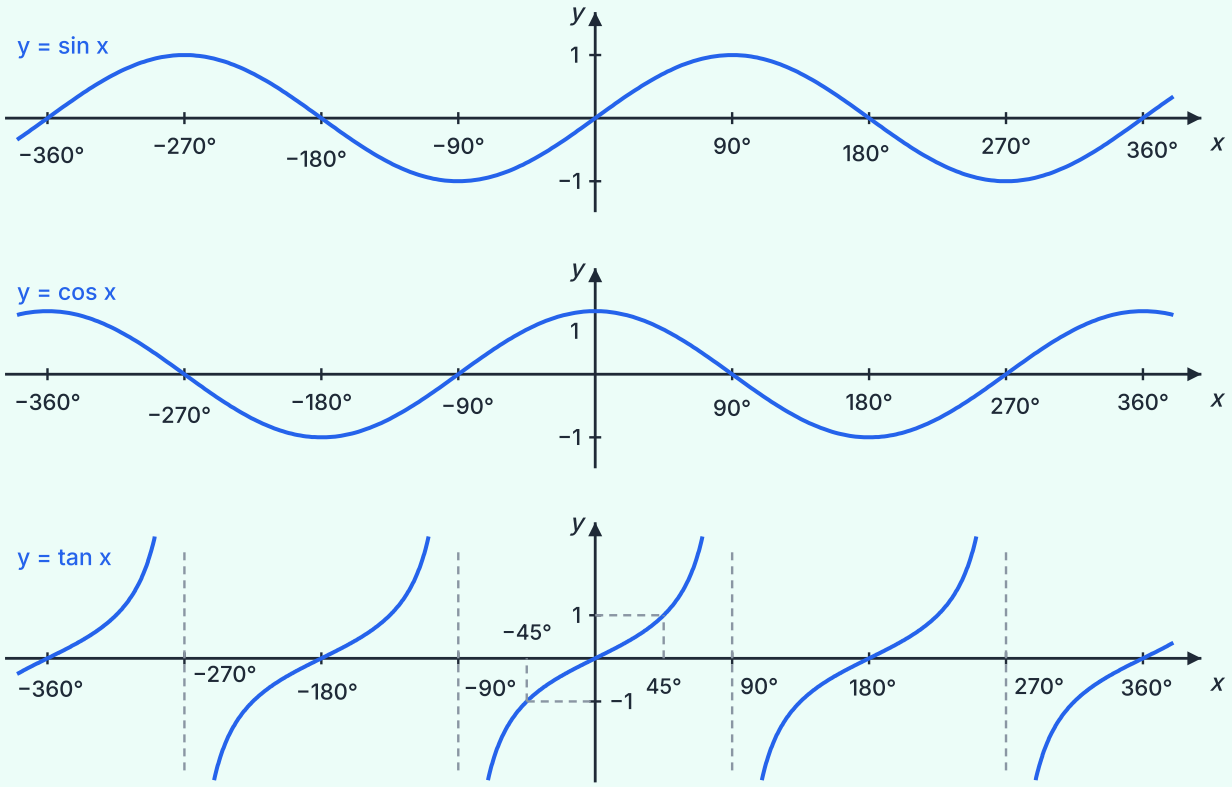
CONCEPT 7 · the graphs of sin, cos and tan

THE GRAPHS OF THE TRIGONOMETRIC FUNCTIONS (YEAR 11 ADVANCED, 5E)

The sine and cosine graphs are waves, and they are the basis of all the mathematics that deals with waves. The later trigonometry in this course will mostly deal with these wave properties. These two graphs each repeat every 360°, and the graphs are therefore said to have a *period* of 360°.

The piece of the sine wave from $\theta = 0°$ to $\theta = 90°$ is enough to construct the whole sine wave and the whole cosine wave — use reflections, rotations and translations.

THE GRAPHS FROM −360° TO 360° (YEAR 11 ADVANCED, 5E)



GRAPHS OF TRIGONOMETRIC FUNCTIONS (MATHS PATHWAY, *GRAPHS OF TRIGONOMETRIC FUNCTIONS*)

Graphing $\sin\theta$, $\cos\theta$ and $\tan\theta$ makes it easier to look at angles which are equivalent on the unit circle. The horizontal axis will show the value of the angle, θ. The vertical axis will show the function value: $\sin\theta$, $\cos\theta$ or $\tan\theta$.

READING THE THREE GRAPHS

- Each graph of sin and cos stays between −1 and 1. Its value at an angle is the y-coordinate (sin) or x-coordinate (cos) of the point on the unit circle.
- The graph of $y = \tan x$ repeats every 180°, so its period is 180°. It has vertical asymptotes at $x = \ldots, -270°, -90°, 90°, 270°, \ldots$, where $\tan x$ is undefined (Box 12).
- Each graph is above the x-axis exactly in the quadrants where ASTC says the function is positive. For example, $\cos x < 0$ for $90° < x < 270°$, quadrants 2 and 3.
- A horizontal line $y = k$ with $0 < k < 1$ or $-1 < k < 0$ meets the sine graph twice between 0° and 360°: two angles with the same sine. The line $y = 0$ meets it three times, at 0°, 180° and 360°.

TUTOR DEMO **Reading the cosine graph**

Use the graph of $y = \cos x$ for $0° \leq x \leq 360°$ to:

a find the values of x for which $\cos x = 0$,

b find the values of x for which $\cos x$ is negative,

c find the other angle between 0° and 360° with the same cosine as 50°.

YOUR TURN **Same method**

Use the graph of $y = \sin x$ for $0° \leq x \leq 360°$ to:

a find the values of x for which $\sin x = 0$,

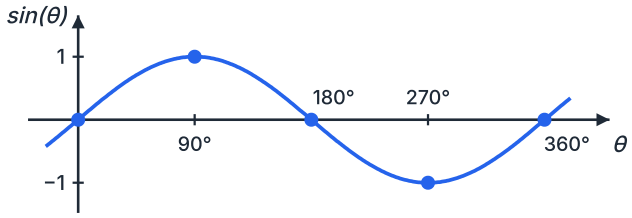
b find the values of x for which $\sin x$ is negative,

c find the other angle between 0° and 360° with the same sine as 50°.

WORKED EXAMPLE

From the text: Maths Pathway, *Graphs of Trigonometric Functions* Q6 and Q7

6. The following graph shows $\sin\theta$ from 0° to 360°.



c) Copy and complete the table below using the values from the graph.

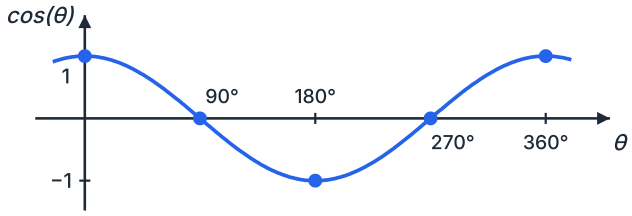
d) Based on the graph, how many angles between 0° and 360° have a sine value of 0.75?

SOLUTION

θ	0°	90°	180°	270°	360°
$\sin\theta$	0	1	0	−1	0

d) 2

7. The following graph shows $\cos\theta$ from 0° to 360°.



c) Copy and complete the table using the values from the graph.

d) Based on the graph, how many angles between 0° and 360° have a cosine value of −0.75?

θ	0°	90°	180°	270°	360°
$\cos\theta$	1	0	−1	0	1

d) 2

STEPS THE SOURCE SKIPS: Q6(D) AND Q7(D)

Draw the horizontal line $y = 0.75$ across the sine graph. It meets the graph once in the first quadrant (rising) and once in the second quadrant (falling), so there are 2 angles. The line $y = -0.75$ meets the cosine graph once in quadrant 2 and once in quadrant 3, where cos is negative.

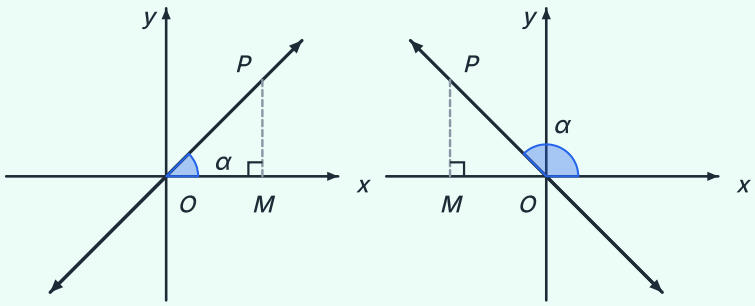
CONCEPT 8 · gradient and the angle of inclination

GRADIENT (YEAR 11 ADVANCED, 6B BOX 5, IN BRIEF)

The gradient of the interval from $P(x_1, y_1)$ to $Q(x_2, y_2)$ is $\dfrac{\text{rise}}{\text{run}} = \dfrac{y_2 - y_1}{x_2 - x_1}$. Horizontal intervals have gradient zero, and vertical intervals have no gradient. A line that rises from left to right has positive gradient, and one that falls has negative gradient.

GRADIENT AND THE ANGLE OF INCLINATION (YEAR 11 ADVANCED, 6B)

The *angle of inclination* of a line is the angle between the upward direction of the line and the positive direction of the x-axis.



The two diagrams above show that lines with positive gradients have acute angles of inclination, and lines with negative gradients have obtuse angles of inclination. They also illustrate the trigonometric relationship between the gradient and the angle of inclination α.

8 ANGLE OF INCLINATION

Suppose that a line has angle of inclination α. Then:

- gradient of line $= \tan\alpha$ OR the line is vertical and $\alpha = 90°$.

PROOF

A When α is acute, as in the first diagram above, then the rise MP and the run OM are the opposite and adjacent sides of the triangle POM, so

$$\tan\alpha = \frac{MP}{OM} = \text{gradient of } OP.$$

B When α is obtuse, as in the second diagram, then $\angle POM = 180° - \alpha$, so

$$\tan\alpha = -\tan\angle POM = -\frac{MP}{OM} = \text{gradient of } OP.$$

STEPS THE BOOK SKIPS: THE PROOF, PART B

$\tan\alpha = -\tan(180° - \alpha)$ is the obtuse-angle rule from Concept 6. In the second diagram M is to the left of O, so the run from O to P is $-OM$, and the gradient of OP is $\dfrac{MP}{-OM} = -\dfrac{MP}{OM}$.

TUTOR DEMO **Angles of inclination**

a Given the points $A(-2, 4)$, $B(-4, 0)$ and $O(0, 0)$, find the angles of inclination of the intervals AB and AO, correct to the nearest degree.

b What sort of triangle is ΔABO?

YOUR TURN **Same method**

a Given the points $A(-4, 3)$, $B(-8, 0)$ and $O(0, 0)$, find the angles of inclination of the intervals AB and AO, correct to the nearest degree.

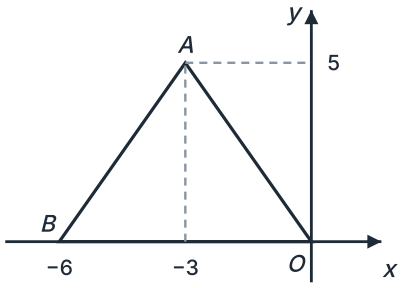
b What sort of triangle is ΔABO?

WORKED EXAMPLE 6 **From the text: keep for homework**

a Given the points $A(-3, 5)$, $B(-6, 0)$ and $O(0, 0)$, find the angles of inclination of the intervals AB and AO.

b What sort of triangle is ΔABO?

SOLUTION



a First, $\text{gradient of } AB = \dfrac{0-5}{-6+3}$

$= \frac{5}{3}$,

and using a calculator to solve $\tan\alpha = \frac{5}{3}$,

angle of inclination of $AB \doteqdot 59°$.

Secondly, $\text{gradient of } AO = \dfrac{0-5}{0+3}$

$= -\frac{5}{3}$,

and using a calculator to solve $\tan\alpha = -\frac{5}{3}$,

angle of inclination of $AO \doteqdot 121°$.

b Hence $\angle AOB = 59°$ (straight angle).
Thus the base angles of ΔAOB are equal, so the triangle is isosceles.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 6

a $\tan^{-1}\frac{5}{3} \doteqdot 59.04°$. For $\tan\alpha = -\frac{5}{3}$ the calculator gives $\tan^{-1}\left(-\frac{5}{3}\right) \doteqdot -59.04°$, which is not an angle of inclination. The gradient is negative, so α is obtuse, and $\tan\alpha = -\tan(180° - \alpha)$ gives $180° - \alpha \doteqdot 59.04°$, so $\alpha \doteqdot 120.96° \doteqdot 121°$.

b $\angle AOB$ and the angle of inclination of AO lie on the straight line BO extended, so they add to 180°: $\angle AOB \doteqdot 180° - 121° = 59°$. The angle of inclination of AB is $\angle ABO \doteqdot 59°$. So $\angle ABO = \angle AOB$.

HOMEWORK · four bands, easy to past-paper level

The bands get harder: **Easy** and **Medium** practise each concept, **Hard** mixes ideas and includes Year 11 Challenge questions, and the **past-paper mini exam** is done last, under exam conditions. Each question names its source. A calculator may be used unless a question asks for an exact value.

EASY

1 Maths Pathway, *Graphs of Trigonometric Functions*, Q1

Sketch each of the following angles on the unit circle. State which angle between 0° and 360° it is equivalent to.

a) $-90°$ b) $-30°$ c) $375°$

d) $-195°$ e) $430°$ f) $720°$

2 Maths Pathway, *Graphs of Trigonometric Functions*, Q3

State which quadrant the following angles belong to.

a) $-45°$ b) $400°$ c) $-185°$ d) $640°$

e) $560°$ f) $-400°$ g) $-98°$ h) $725°$

3 Maths Pathway, *The Unit Circle*, Q9

State whether each of the following values will be positive or negative.

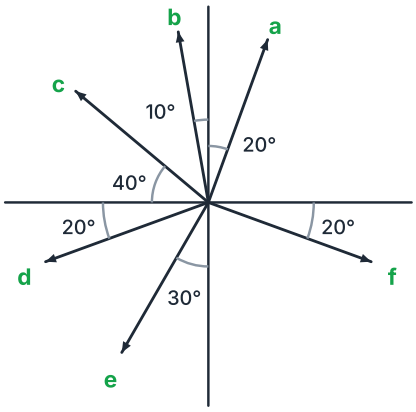
Hint: sketch a diagram if you get stuck.

a) $\sin(26°)$ b) $\cos(26°)$ c) $\tan(26°)$ d) $\sin(331°)$

e) $\cos(162°)$ f) $\sin(179°)$ g) $\cos(223°)$ h) $\tan(91°)$

4 Year 11 Advanced Exercise 5D Q5

Write down two positive angles between 0° and 720° and two negative angles between −720° and 0° that are represented by each of the rays in the diagram to the right.



5 Year 11 Advanced Exercise 5E Q1

Use the ASTC rule to determine the sign (+ or −) of each trigonometric ratio.

a $\sin 20°$	b $\cos 50°$	c $\cos 100°$	d $\tan 140°$
e $\tan 250°$	f $\sin 310°$	g $\sin 200°$	h $\cos 280°$
i $\sin 340°$	j $\cos 350°$	k $\tan 290°$	l $\cos 190°$
m $\tan 170°$	n $\sin 110°$	o $\tan 80°$	p $\cos 170°$

6 Year 11 Advanced Exercise 5E Q2

Find the related acute angle of each angle.

a $10°$	b $150°$	c $310°$	d $200°$	e $80°$
f $250°$	g $290°$	h $100°$	i $350°$	j $160°$

7 Year 11 Advanced Exercise 5E Q3

Write each trigonometric ratio as the ratio of an acute angle with the correct sign attached.

a $\tan 130°$	b $\cos 310°$	c $\sin 220°$	d $\tan 260°$
e $\cos 170°$	f $\sin 320°$	g $\cos 185°$	h $\sin 125°$
i $\tan 325°$	j $\sin 85°$	k $\cos 95°$	l $\tan 205°$

8 Year 11 Advanced Exercise 5E Q4

Use the trigonometric graphs to find the values (if they exist) of these trigonometric ratios of boundary angles.

a $\sin 0°$	b $\cos 180°$	c $\cos 90°$	d $\tan 0°$
e $\sin 90°$	f $\cos 0°$	g $\sin 270°$	h $\tan 270°$
i $\sin 180°$	j $\cos 270°$	k $\tan 90°$	l $\tan 180°$

9 Year 11 Advanced Exercise 5E Q5

Find the exact value of:

a $\sin 60°$	b $\sin 120°$	c $\sin 240°$	d $\sin 300°$
e $\cos 45°$	f $\cos 135°$	g $\cos 225°$	h $\cos 315°$
i $\tan 30°$	j $\tan 150°$	k $\tan 210°$	l $\tan 330°$

10 Year 11 Advanced Exercise 6B Q6

Use the formula gradient $= \tan \alpha$ to find the gradient, correct to two decimal places where necessary, of a line with angle of inclination:

a $15°$	b $135°$	c $22\frac{1}{2}°$	d $72°$

11 Year 11 Advanced Exercise 6B Q7

Use the formula gradient $= \tan\alpha$ to find the angle of inclination, correct to the nearest degree where necessary, of a line with gradient:

a 1 b $-\sqrt{3}$ c 4 d $\dfrac{1}{\sqrt{3}}$

12 Year 11 Advanced Exercise 6B Q8

The gradient measures the steepness of a line. Each of the following lines AB has positive gradient. By comparing the gradient to 1, determine if the angle of inclination is less than, equal to, or more than 45°.

a $A(1,3),\ B(6,5)$

b $A(4,1),\ B(7,4)$

c $A(-3,2),\ B(1,5)$

d $A(-3,4),\ B(-7,-2)$

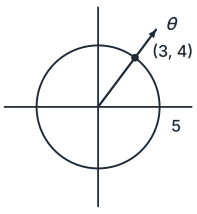
MEDIUM

13 Year 11 Advanced Exercise 5D Q6

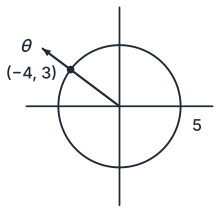
Use the definitions $\sin\theta = \dfrac{y}{r}$ and $\cos\theta = \dfrac{x}{r}$ and $\tan\theta = \dfrac{y}{x}$ to write down the values of the six trigonometric ratios of the angle θ in each diagram.

FOR THIS PACK Write down $\sin\theta$, $\cos\theta$ and $\tan\theta$ only (see the note on cosec, sec and cot in Concept 2).

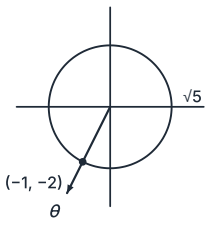
a



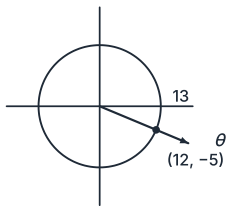
b



c



d



14 Year 11 Advanced Exercise 5E Q6

Find the exact value of:

a $\cos 120°$ b $\tan 225°$ c $\sin 330°$ d $\sin 135°$

e $\tan 240°$ f $\cos 210°$ g $\tan 315°$ h $\cos 300°$

i $\sin 225°$ j $\cos 150°$ k $\sin 210°$ l $\tan 300°$

15 Year 11 Advanced Exercise 5E Q9

Find the related acute angle of each angle.

a $-60°$ b $-200°$ c $-150°$ d $-300°$

e $430°$ f $530°$ g $590°$ h $680°$

16 Year 11 Advanced Exercise 5E Q10

Find the exact value of:

a $\cos(-60°)$ b $\sin(-120°)$ c $\tan(-120°)$

d $\sin(-315°)$ e $\tan(-210°)$ f $\cos(-225°)$

g $\tan 420°$ h $\cos 510°$ i $\sin 495°$

j $\sin 690°$ k $\cos 600°$ l $\tan 585°$

17 Maths Pathway, *Using Symmetry on the Unit Circle*, Q9

Use the following table to find the missing values below.

θ	15°	30°	45°	60°
$\sin\theta$	0.26	0.5	0.71	0.87
$\cos\theta$	0.97	0.87	0.71	0.5
$\tan\theta$	0.27	0.58	1	1.73

a) $\sin(165°)$ b) $\cos(165°)$ c) $\tan(315°)$ d) $\sin(210°)$

e) $\cos(195°)$ f) $\tan(240°)$ g) $\sin(300°)$ h) $\cos(330°)$

18 Maths Pathway, *Graphs of Trigonometric Functions*, Q5

The following questions look at equivalent angles with the same sine and cosine values.
Hint: you can use your sketches from question 2.

a) Which angles between 360° and 540° have the same sine value as 30°?

b) Which angles between −180° and 180° have the same sine value as 330°?

c) Which angles between 450° and 630° have the same cosine value as 95°?

d) Which angles between 360° and 720° have the same cosine value as 170°?

ABOUT THE HINT Question 2 of the sheet asks for unit circles labelled with the angle at the start of each quadrant, for 0° to 360°, 360° to 720°, −360° to 0° and 720° to 1080°.

19 Maths Pathway, *Graphs of Trigonometric Functions*, Q9

The following questions will be looking at trigonometric functions between −180° and 180°.

a) Copy and complete the following table.

θ	−180°	−90°	0°	90°	180°
$\sin\theta$					
$\cos\theta$					
$\tan\theta$					

b) Use your table of values from part a) to graph $\sin\theta$, $\cos\theta$ and $\tan\theta$ from −180° to 180°.

- Each graph should be on its own set of axes
- Label the horizontal axes from −180° to 180°, marking every 90 degrees
- Label the vertical axes from −1.5 to 1.5, marking every 0.5 units.

20 Year 11 Advanced Exercise 6B Q17

Find the points A and B where each line below meets the x-axis and y-axis respectively. Hence find the gradient of AB and its angle of inclination α (correct to the nearest degree).

a $y = 3x + 6$

b $y = -\frac{1}{2}x + 1$

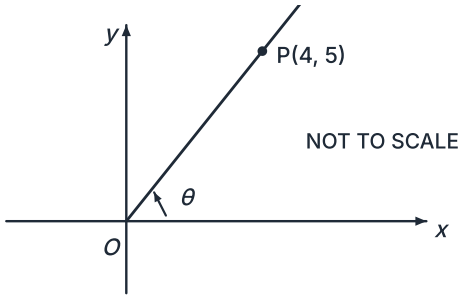
c $3x + 4y + 12 = 0$

d $\frac{x}{3} - \frac{y}{2} = 1$

21 Past paper: Northern Beaches Christian School, 2022, Mathematics Year 10 5.3/Year 9 Accelerated Yearly Examination, Section I Q13 (1 mark)

The diagram shows the interval OP where O is the origin and P has coordinates $(4, 5)$.

What is the size (to the nearest degree) of the angle represented by θ in the diagram?



(A) 38°
(B) 39°
(C) 51°
(D) 52°

Hint: What do you know about the diagram?

22 Past paper: Sydney Grammar School, 2020, Annual Examination Form IV Mathematics, Q6(c) (3 marks)

(i) Sketch the graph of $y = \cos\theta$ for $0° \leq \theta \leq 360°$. 2

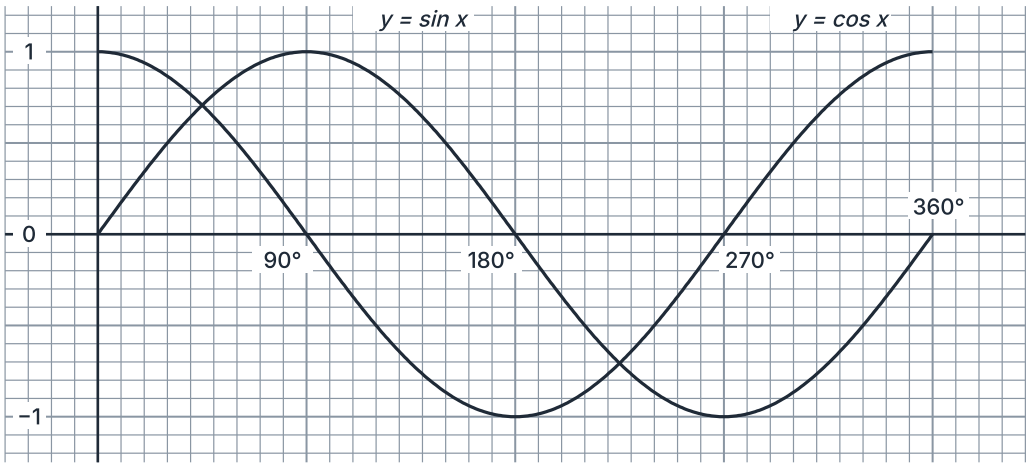
(ii) Hence solve $\cos\theta = 0$ for $0° \leq \theta \leq 360°$. 1

Hint: What do you know about the graph of $y = \cos\theta$?

HARD

23 Year 11 Advanced Exercise 5D Q8

Use the following diagram to answer the questions.



a Read off the diagram, correct to two decimal places where necessary, the values of:

i $\cos 60°$ ii $\sin 210°$ iii $\sin 72°$ iv $\cos 18°$ v $\sin 144°$

vi $\cos 36°$ vii $\cos 153°$ viii $\sin 27°$ ix $\sin 234°$ x $\cos 306°$

b Find from the graphs two values of x between 0° and 360° for which:

i $\sin x = 0.5$ ii $\cos x = -0.5$ iii $\sin x = 0.9$ iv $\cos x = 0.6$

v $\sin x = 0.8$ vi $\cos x = -0.8$ vii $\sin x = -0.4$ viii $\cos x = -0.3$

c Find two values of x between 0° and 360° for which $\sin x = \cos x$.

24 Year 11 Advanced Exercise 5E Q12

Given that $\sin 25° \fallingdotseq 0.42$ and $\cos 25° \fallingdotseq 0.91$, write down approximate values, without using a calculator, for:

a $\sin 155°$ b $\cos 205°$ c $\cos 335°$

d $\sin 335°$ e $\sin 205° - \cos 155°$ f $\cos 385° - \sin 515°$

25 Year 11 Advanced Exercise 5E Q14

Show by substitution into LHS and RHS that each trigonometric identity is satisfied by the given values of the angles.

a Show that $\sin 2\theta = 2\sin\theta\cos\theta$, when $\theta = 150°$.

b Show that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, when $\theta = 225°$.

c Show that $\sin(A + B) = \sin A\cos B + \cos A\sin B$, when $A = 300°$ and $B = 240°$.

26 Year 11 Advanced Exercise 5E Q15 (part d omitted: it uses sec)

Write as a trigonometric ratio of θ, with the correct sign attached:

a $\sin(-\theta)$ b $\cos(-\theta)$ c $\tan(-\theta)$ e $\sin(180° - \theta)$

f $\sin(360° - \theta)$ g $\cos(180° - \theta)$ h $\tan(180° + \theta)$

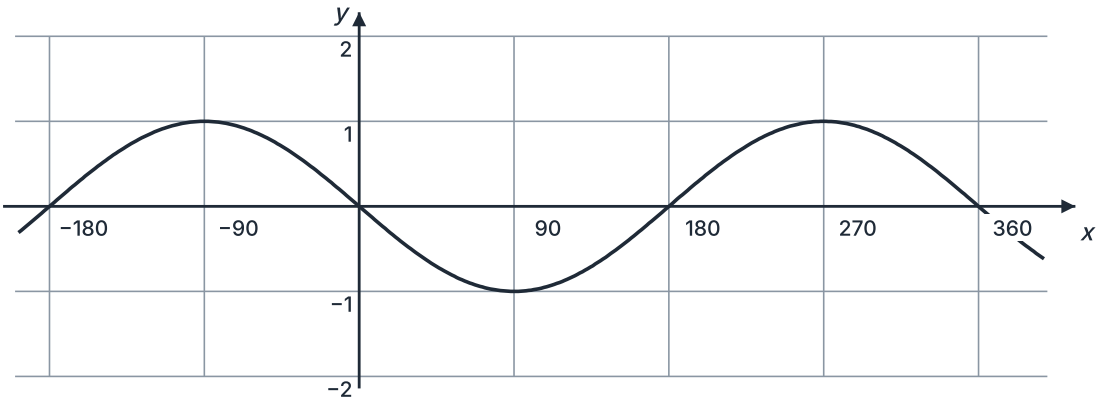
27 Year 11 Advanced Exercise 6B Q20

Find the gradient, correct to two decimal places where appropriate, of a line sloping upwards, if its acute angle with the y-axis is:

a $15°$ b $45°$ c $22\frac{1}{2}°$ d $72°$

28 Past paper: Sydney Boys High School, 2020, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics Stage 5.3, Q8 (1 mark)

What is the equation of the graph below?



A. $y = \sin x°$ B. $y = \sin(-x)°$ C. $y = \cos x°$ D. $y = -\cos x°$

Hint: What can you say about the graph?

PAST-PAPER HOMEWORK · mini exam

Suggested time: 25 minutes **17 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Sydney Boys High School, 2020, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics Stage 5.3, Q4** **1 mark**

Given that $\sin\theta > 0$ and $\cos\theta < 0$, which quadrant does θ lie in?

A. Quadrant 1
B. Quadrant 2
C. Quadrant 3
D. Quadrant 4

Hint: What does the question give you?

PP 2 **Girraween High School, 2022, Year 10 Yearly Examination Mathematics, Part A, Q12** **1 mark**

The obtuse angle θ such that $\tan\theta = -\tan 50°$ is

(A) $\theta = 130°$ (B) $\theta = 140°$ (C) $\theta = 145°$ (D) $\theta = 155°$

Hint: What do you know about $\tan 50°$ and obtuse angles?

PP 3 **Girraween High School, 2022, Year 10 Yearly Examination Mathematics, Part A, Q13** **1 mark**

The incorrect statement below is

(A) $\cos\theta > 0,\ \sin\theta < 0 \Rightarrow \theta$ is in the fourth quadrant.
(B) $\cos\theta > 0,\ \tan\theta > 0 \Rightarrow \theta$ is in the third quadrant.
(C) $\tan\theta < 0,\ \cos\theta < 0 \Rightarrow \theta$ is in the second quadrant.
(D) $\sin\theta > 0,\ \tan\theta > 0 \Rightarrow \theta$ is in the first quadrant.

Hint: What do you know about each statement?

PP 4 **Girraween High School, 2023, Year 10 Mathematics Yearly Exam, Multiple Choice, Q3** 1 mark

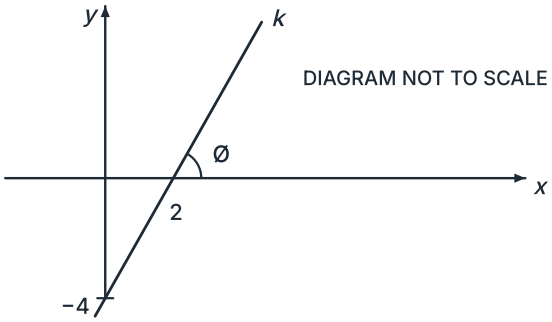
To the nearest degree, what is the size of angle Ø shown in this diagram?

A) 26°

B) 27°

C) 63°

D) 64°



Hint: What does the diagram tell you about line k?

PP 5 **The Scots College, 2024, Year 10 Assessment Task 4, Mathematics 5.3, Q28(a)** 1 mark

If θ is acute, find the value of θ in $\cos\theta = \cos 286°$

Hint: What do you know about $\cos\theta$ and the angle 286°?

PP 6 **Sydney Boys High School, 2020, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics Stage 5.3, Q14** 1 mark

Find the angle of inclination that the line $y = 4x - 5$ makes with the positive direction of the x axis.

Correct your answer to the nearest minute.

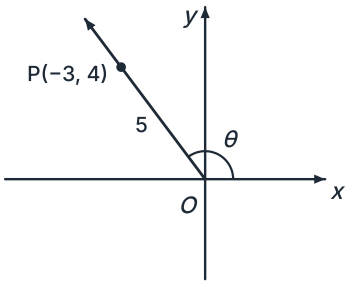
Hint: What do you know about the line $y = 4x - 5$?

PP 7 **Sydney Grammar School, 2020, Annual Examination Form IV Mathematics, Q1(h)** 2 marks

Draw a quadrant diagram to represent 480° and state its related angle.

Hint: What do you know about the angle 480°?

PP 8 **Sydney Boys High School, 2021, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part C, (f)** 2 marks



In the diagram above, the ray OP passes through the point $P(-3, 4)$ and makes the angle θ with the positive x-axis.
Given $OP = 5$ units, write down the values of $\sin\theta$ and $\tan\theta$.

Hint: What does the diagram give you?

PP 9 **North Sydney Girls High School, 2022, Year 10 Mathematics Yearly Examination, Q16(c)** 2 marks

Draw a neat sketch of the graph of $y = \tan x$ for $0° \leq x \leq 180°$.

Hint: What do you know about the graph of $y = \tan x$?

PP 10 **North Sydney Boys High School, 2020, Stage 5 Mathematics Year 10 Yearly Examination, Q2(b)** 2 marks

Find the exact value of $\dfrac{\sin 150° + \cos 300°}{\tan 750° - \tan 120°}$

Hint: What do you notice about the angles in the expression?

PP 11 **Sydney Technical High School, 2020, Year 10 Mathematics Assessment 4, Q7(c)** 3 marks

Find the exact value of

(i) $\cos 45°$

(ii) $\sin 240°$

(iii) $\tan^2(420°)$

Hint: What do you know about each of these angles?

LESSON T2

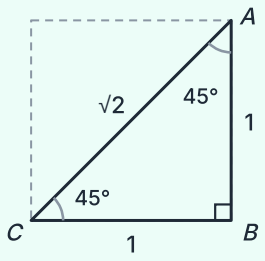
Exact values and trigonometric equations

The exact values of sin, cos and tan at 30°, 45° and 60° come from two special triangles. With the complementary-angle identity and the ratio identity, they give exact answers without a calculator. Given one trigonometric function, a circle diagram and Pythagoras' theorem give the others. Then trigonometric equations are solved over a given domain by the quadrants-and-related-angle method, and the Hard band adds compound angles such as $2x$. Sources: Year 11 Advanced book 5A, 5F, 5G and 5H (and Example 22 of 5I); Maths Pathway sheet *Solving Trigonometric Equations*. All angles are in degrees.

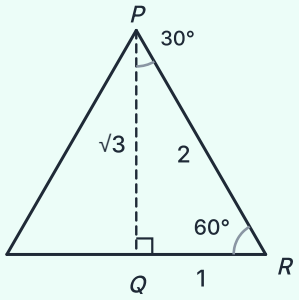
CONCEPT 1 · exact values for 30°, 45° and 60°

SPECIAL ANGLES (YEAR 11 ADVANCED, 5A)

The values of the trigonometric functions for the three acute angles 30°, 45° and 60° can be calculated exactly using half a square and half an equilateral triangle, and applying Pythagoras' theorem.



Take half a square with side length 1.
The resulting right-angled triangle ABC has two angles of 45°.
By Pythagoras' theorem, the hypotenuse AC has length $\sqrt{2}$



Take half an equilateral triangle with side length 2 by dropping an altitude.
The resulting right-angled ΔPQR has angles of 60° and 30°.
By Pythagoras' theorem, $PQ = \sqrt{3}$.

Applying the definitions in Box 1 gives the values in the table below.

2 A TABLE OF EXACT VALUES

θ	$\sin\theta$	$\cos\theta$	$\tan\theta$
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$

TRIGONOMETRIC FUNCTIONS OF OTHER ANGLES (YEAR 11 ADVANCED, 5A)

A calculator is usually used to approximate trigonometric functions of other angles. Always check first that the calculator is in degrees mode — there is usually a key labelled mode or DRG or something similar.

READING THE TABLE FROM THE TRIANGLES

Box 1 is $\sin\theta = \dfrac{\text{opp}}{\text{hyp}}$, $\cos\theta = \dfrac{\text{adj}}{\text{hyp}}$, $\tan\theta = \dfrac{\text{opp}}{\text{adj}}$ (see the revision pages). In ΔABC, from the 45° angle at C: opp = 1, adj = 1, hyp = $\sqrt{2}$. In ΔPQR, from the 60° angle at R: opp = $\sqrt{3}$, adj = 1, hyp = 2; from the 30° angle at P: opp = 1, adj = $\sqrt{3}$, hyp = 2. The book's table also has cosec, sec and cot columns, which are not in the Year 10 course and are left out. The book's next sentence, on radian mode, is also left out: every angle in this pack is in degrees.

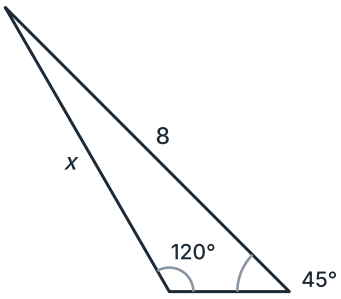
Exact values are often written with a rational denominator: $\dfrac{1}{\sqrt{2}} = \dfrac{\sqrt{2}}{2}$ and $\dfrac{1}{\sqrt{3}} = \dfrac{\sqrt{3}}{3}$. Combined with Lesson T1, this gives exact values for every angle whose related angle is 30°, 45° or 60°.

TUTOR DEMO **Exact values**

a Find the exact value of $\cos^2 45° + \sin 30°$.

b Find the exact value of $\tan 30° \times \tan 60°$.

c Use the sine rule and exact values to find the side x in the triangle shown, in simplest exact form.

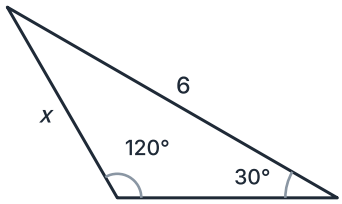


YOUR TURN **Same method**

a Find the exact value of $\sin^2 60° + \cos 60°$.

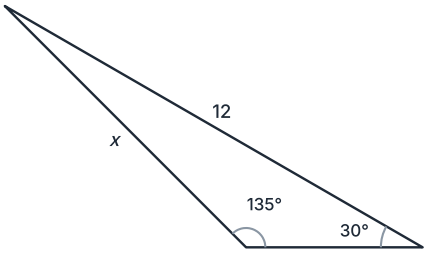
b Find the exact value of $\cos 30° \times \tan 30°$.

c Use the sine rule and exact values to find the side x in the triangle shown, in simplest exact form.



WORKED EXAMPLE 22 **From the text (5I): keep for homework**

Find the side x in the triangle shown to the right.



SOLUTION

Using the sine rule, and placing the unknown at the top left,

$$\frac{x}{\sin 30°} = \frac{12}{\sin 135°}$$

$$x = \frac{12 \sin 30°}{\sin 135°}. \qquad (\times \sin 30°)$$

Using special angles, $\sin 30° = \frac{1}{2}$,

and $\sin 135° = +\sin 45° \qquad (\text{sine is positive for obtuse angles})$

$$= \frac{1}{\sqrt{2}}. \qquad (\text{the related acute angle is } 45°)$$

Hence $x = 12 \times \frac{1}{2} \times \frac{\sqrt{2}}{1}$

$$= 6\sqrt{2}.$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 22

Dividing by $\sin 135° = \frac{1}{\sqrt{2}}$ is the same as multiplying by its reciprocal $\frac{\sqrt{2}}{1}$. Then $12 \times \frac{1}{2} = 6$, so $x = 6 \times \sqrt{2} = 6\sqrt{2}$. The sine rule is from Topic 11 (see the revision pages).

CONCEPT 2 · tan = sin/cos and complementary angles

TRIGONOMETRIC IDENTITIES (YEAR 11 ADVANCED, 5G)

Working with the trigonometric functions requires knowledge of a number of formulae called *trigonometric identities*, which relate trigonometric functions to each other.

The two ratio identities. Again using the definitions of the trigonometric functions:

20 THE RATIO IDENTITIES

For any angle θ: $\tan\theta = \frac{\sin\theta}{\cos\theta}$ (provided that $\cos\theta \neq 0$)

The three identities for complementary angles. The angles θ and $90° - \theta$ are called *complementary angles* because they add to a right angle. Three trigonometric identities relate the values of the trigonometric functions at an angle θ and the complementary angle $90° - \theta$.

22 THE COMPLEMENTARY ANGLE IDENTITIES

For any angle θ: $\cos(90° - \theta) = \sin\theta$

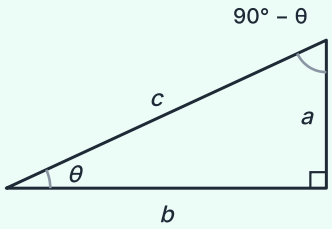
For example, $\cos 20° = \sin 70°$.

PROOF

A [Acute angles]

The triangle to the right shows that when a right-angled triangle is viewed from $90° - \theta$ instead of from θ, then the opposite side and the adjacent side are exchanged. Hence

$$\cos(90° - \theta) = \frac{a}{c} = \sin\theta.$$



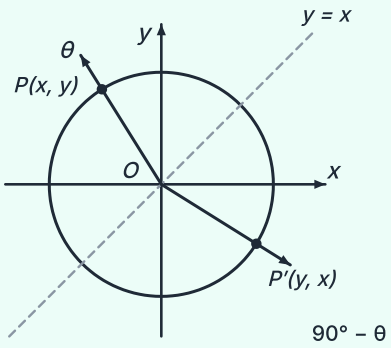
B [General angles]

For general angles, we take the full circle diagram, and reflect it in the diagonal line $y = x$. Let P' be the image of P under this reflection.

1. The image OP' of the ray OP corresponds to the angle $90° - \theta$.
2. The image P' of $P(x, y)$ has coordinates $P'(y, x)$, because reflection in the line $y = x$ reverses the coordinates of each point.

Applying the definitions of the trigonometric functions to the angle $90° - \theta$:

$$\cos(90° - \theta) = \frac{y}{r} = \sin\theta.$$



THE OTHER COMPLEMENTARY IDENTITIES

Box 20 also has $\cot\theta = \dfrac{\cos\theta}{\sin\theta}$, and Box 22 and its proof also have lines for cot and cosec. They are not in the Year 10 course and are left out.

In words, the sine of an angle is the cosine of its complement, and the cosine of an angle is the sine of its complement: $\sin A = \cos(90° - A)$ and $\cos A = \sin(90° - A)$. This is where the name *co*sine comes from (5G Box 23: cosine θ = sine (complement of θ)).

TUTOR DEMO **Complementary angles and tan = sin/cos**

a Find the value of x if $\sin 40° = \cos x$, where $0° < x < 90°$.

b Simplify $\dfrac{\cos(90° - \theta)}{\cos \theta}$.

c Without a calculator, find the value of $\sin 25° - \cos 65°$.

YOUR TURN **Same method**

a Find the value of x if $\cos 15° = \sin x$, where $0° < x < 90°$.

b Simplify $\sin(90° - \theta) \tan \theta$.

c Without a calculator, find the value of $\cos 32° - \sin 58°$.

NO WORKED EXAMPLE IN THE BOOK HERE

Section 5G has no worked example on complementary angles; its examples prove identities that use cosec, sec, cot or $\sin^2 \theta + \cos^2 \theta = 1$, which are outside the Year 10 course. Use the example in Box 22 and the tutor demo as your model.

CONCEPT 3 · given one trigonometric function, find another

GIVEN ONE TRIGONOMETRIC FUNCTION, FIND ANOTHER (YEAR 11 ADVANCED, 5F)

When the exact value of one trigonometric function is known for an angle, the exact values of the other trigonometric functions can easily be found using the circle diagram and Pythagoras' theorem.

18 GIVEN ONE TRIGONOMETRIC FUNCTION, FIND ANOTHER

- Place a ray or rays on a circle diagram in the quadrants allowed in the question.
- Complete the triangle and use Pythagoras' theorem to find whichever of x, y and r is missing.

HOW THE DEFINITIONS ARE USED

From Lesson T1, $\sin\theta = \frac{y}{r}$, $\cos\theta = \frac{x}{r}$ and $\tan\theta = \frac{y}{x}$, with $r > 0$ always. The given ratio tells you two of x, y and r (any circle will do, because only the ratios matter), Pythagoras' theorem $x^2 + y^2 = r^2$ gives the third, and the quadrant gives its sign.

TUTOR DEMO One ratio gives the others

It is known that $\cos\theta = \frac{2}{7}$.

a Find the possible values of $\sin\theta$.

b Find $\sin\theta$ if it is also known that $\tan\theta$ is negative.

YOUR TURN Same method

It is known that $\cos\theta = -\frac{1}{3}$.

a Find the possible values of $\sin\theta$.

b Find $\sin\theta$ if it is also known that $\tan\theta$ is positive.

WORKED EXAMPLE 11 **From the text: keep for homework**

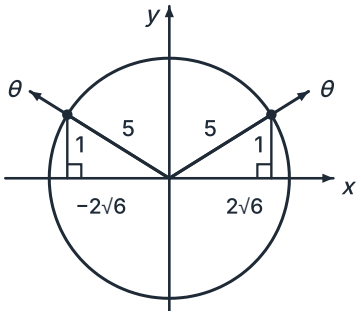
It is known that $\sin\theta = \frac{1}{5}$.

a Find the possible values of $\cos\theta$.

b Find $\cos\theta$ if it is also known that $\tan\theta$ is negative.

SOLUTION

a The angle must be in quadrant 1 or 2, because $\sin\theta$ is positive.



Because $\sin\theta = \frac{y}{r} = \frac{1}{5}$, we can take $y = 1$ and $r = 5$,

so by Pythagoras' theorem, $x = \sqrt{24}$ or $-\sqrt{24}$,

so $\cos\theta = \frac{2\sqrt{6}}{5}$ or $-\frac{2\sqrt{6}}{5}$.

b Because $\tan\theta$ is negative, θ can only be in quadrant 2,

so $\cos\theta = -\frac{2\sqrt{6}}{5}$.

Note: In the diagrams of this section, some of the 'side lengths' on the horizontal and vertical sides of triangles are marked as negative because they are really displacements from the origin rather than lengths. Always trust the quadrants to give you the correct sign.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 11

a $x^2 + 1^2 = 5^2$, so $x^2 = 24$ and $x = \pm\sqrt{24} = \pm\sqrt{4 \times 6} = \pm 2\sqrt{6}$. Then $\cos\theta = \frac{x}{r} = \pm\frac{2\sqrt{6}}{5}$. **b** $\tan\theta$ is negative in quadrants 2 and 4, and $\sin\theta$ is positive in quadrants 1 and 2, so the only quadrant allowed is 2, where $x < 0$.

CONCEPT 4 · trigonometric equations: the domain and boundary angles

TRIGONOMETRIC EQUATIONS (YEAR 11 ADVANCED, 5H)

This piece of work is absolutely vital, because so many problems in later work end up with a trigonometric equation that has to be solved.

There are many small details and qualifications in the methods, and the subject needs a great deal of careful study.

Pay attention to the domain. To begin with a warning, before any other details:

25 THE DOMAIN

Always pay careful attention to the domain in which the angle can lie.

Equations involving boundary angles. Boundary angles are a special case because they do not lie in any quadrant.

26 THE BOUNDARY ANGLES

If the solutions are boundary angles, read the solutions off a sketch of the graph.

INVERSE TRIGONOMETRIC FUNCTIONS (MATHS PATHWAY, *SOLVING TRIGONOMETRIC EQUATIONS*)

The inverse of a trigonometric function swaps the input and the output. For example:

Function	Inverse function
$\sin(90°) = 1$	$\sin^{-1}(1) = 90°$
$\cos(0°) = 1$	$\cos^{-1}(1) = 0°$
$\tan(45°) = 1$	$\tan^{-1}(1) = 45°$

If we know the value of $\sin\theta$, $\cos\theta$, or $\tan\theta$, we can use the inverse function to find value of θ.

ONE ANSWER FROM THE CALCULATOR, MANY FROM THE GRAPH

The calculator gives just one angle. An equation such as $\sin x = -1$ has a solution wherever the graph of $y = \sin x$ meets the line $y = -1$, and over a domain of 720° there are two such places. Always count the solutions on a sketch over the whole domain (Box 25).

TUTOR DEMO **Boundary angles**

a Solve $\cos x = 1$, for $0° \leq x \leq 720°$.

b Solve $\cos x = 0$, for $0° \leq x \leq 720°$.

YOUR TURN **Same method**

a Solve $\sin x = 1$, for $0° \leq x \leq 720°$.

b Solve $\tan x = 0$, for $0° \leq x \leq 720°$.

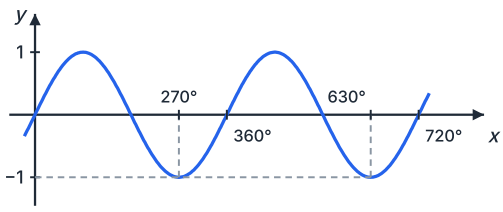
WORKED EXAMPLE 15 **From the text: keep for homework**

a Solve $\sin x = -1$, for $0° \leq x \leq 720°$.

b Solve $\sin x = 0$, for $0° \leq x \leq 720°$.

SOLUTION

a The graph of $y = \sin x$ is shown to the right.
Examine where the curve touches the line $y = -1$, and read off the x-coordinates of these points.
The solution is $x = 270°$ or $630°$.



b Examine where the graph crosses the x-axis.
The solution is $x = 0°, 180°, 360°, 540°$ or $720°$.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 15

The sine graph repeats every 360°, so after the minimum at 270° the next minimum is at $270° + 360° = 630°$. The zeros are every 180°: 0°, 180°, 360°, 540° and 720°, and both ends of the domain count because the domain uses $\leq$.

CONCEPT 5 · the quadrants-and-related-angle method

THE STANDARD METHOD — QUADRANTS AND THE RELATED ACUTE ANGLE (YEAR 11 ADVANCED, 5H)

Most trigonometric equations eventually come down to one or more equations such as

$\sin x = -\frac{1}{2}$, where $-180° \leq x \leq 180°$.

Provided that the angle is not a boundary angle, the method is:

27 THE QUADRANTS-AND-RELATED-ANGLE METHOD

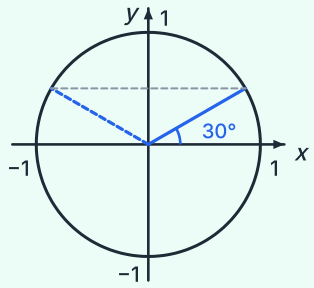
1. Draw a quadrant diagram, then draw a ray in each quadrant that the angle could be in.
2. Find the related acute angle (only work with positive numbers here):
   a using special angles, or
   b using the calculator to find an approximation.
   Never enter a negative number into the calculator at this point.
3. Mark the angles on the ends of the rays, taking account of any restrictions on x, and write a conclusion.

MULTIPLE SOLUTIONS WITH A CALCULATOR (MATHS PATHWAY, *SOLVING TRIGONOMETRIC EQUATIONS*)

Calculators will only give one solution for a trigonometric function, even though there may be multiple solutions. You can use what you know about the unit circle to find any other solutions.

For example, solving the following equation with a calculator gives us a single solution:

$$\sin(x) = \frac{1}{2}$$
$$x = \sin^{-1}\left(\frac{1}{2}\right)$$
$$= 30$$



But looking at the unit circle, we can see ***two*** possible solutions.

So, x is actually equal to $30°$, $150°$, or any other equivalent angles.

TUTOR DEMO **The standard method**

a Solve the equation $\cos x = -\frac{\sqrt{3}}{2}$, for $-180° \leq x \leq 180°$.

b Solve the equation $\sin x = 0.6$, for $0° \leq x \leq 360°$, correct to the nearest degree.

YOUR TURN **Same method**

a Solve the equation $\tan x = -1$, for $-180° \leq x \leq 180°$.

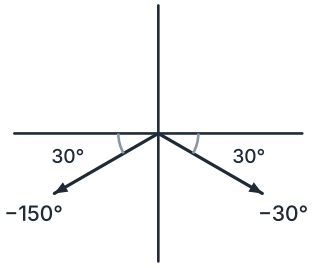
b Solve the equation $\cos x = -0.35$, for $0° \leq x \leq 360°$, correct to the nearest degree.

WORKED EXAMPLE 16 **From the text: keep for homework**

Solve the equation $\sin x = -\frac{1}{2}$, for $-180° \leq x \leq 180°$.

SOLUTION

Here $\sin x = -\frac{1}{2}$, where $-180° \leq x \leq 180°$.
Because $\sin x$ is negative, x is in quadrant 3 or 4.
The sine of the related acute angle is $+\frac{1}{2}$,
so the related angle is 30°.
Hence $x = -150°$ or $-30°$.

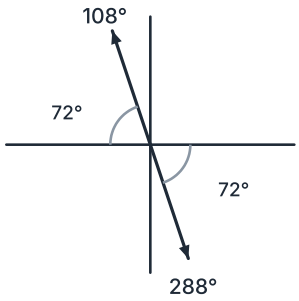


WORKED EXAMPLE 17 **From the text: keep for homework**

Solve the equation $\tan x = -3$, for $0° \leq x \leq 360°$, correct to the nearest degree.

SOLUTION

Here $\tan x = -3$, where $0° \leq x \leq 360°$.
Because $\tan x$ is negative, x is in quadrant 2 or 4.
The tangent of the related acute angle is $+3$,
so the related angle is about 72°.
Hence $x \doteqdot 108°$ or $288°$.



Note: When using a calculator, *never enter a negative number and take an inverse trigonometric function of it.*

In the example above, the calculator was used to find the *related acute angle* whose tan was 3, which is $71°34'$, correct to the nearest minute. The positive number 3 was entered, not −3.

STEPS THE BOOK SKIPS: WORKED EXAMPLES 16 AND 17

16 In the domain $-180° \leq x \leq 180°$, quadrant 3 is reached by turning clockwise: the ray 30° below the negative x-axis is $-(180° - 30°) = -150°$, and the ray in quadrant 4 is $-30°$.

17 $\tan^{-1} 3 \fallingdotseq 71.57° \fallingdotseq 72°$. Quadrant 2: $180° - 71.57° \fallingdotseq 108.43° \fallingdotseq 108°$. Quadrant 4: $360° - 71.57° \fallingdotseq 288.43° \fallingdotseq 288°$. Round only at the end.

LEFT OUT: BOX 28 AND EXAMPLE 18

Box 28 and Example 18 solve equations in cosec and sec by taking reciprocals. These functions are not in the Year 10 course.

CONCEPT 6 · equations with more than one trigonometric function

EQUATIONS WITH MORE THAN ONE TRIGONOMETRIC FUNCTION (YEAR 11 ADVANCED, 5H)

Some trigonometric equations involve more than one trigonometric function. For example,

$$\sin x + \sqrt{3} \cos x = 0$$

The general approach is to use trigonometric identities to produce an equation in only one trigonometric function.

TUTOR DEMO **Divide by cos x**

Solve $\sin x = 2 \cos x$, where $0° \leq x \leq 360°$, correct to the nearest degree.

YOUR TURN **Same method**

Solve $\sqrt{3}\sin x = \cos x$, where $0° \leq x \leq 360°$.

WORKED EXAMPLE 21 **From the text: keep for homework**

Solve $\sin x + \sqrt{3}\cos x = 0$, where $0° \leq x \leq 360°$.

SOLUTION

$$\sin x + \sqrt{3}\cos x = 0$$

$$\tan x + \sqrt{3} = 0 \qquad (\div \cos x)$$

$$\tan x = -\sqrt{3}, \text{ where } 0° \leq x \leq 360°.$$

Because $\tan x$ is negative, x is in quadrants 2 or 4.

The tan of the related acute angle is $\sqrt{3}$, so the related angle is 60°.

Hence $x = 120°$ or $300°$.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 21

Dividing by $\cos x$ uses Box 20, $\dfrac{\sin x}{\cos x} = \tan x$. It is safe: if $\cos x = 0$, the equation would say $\sin x = 0$ as well, but sin and cos are never both 0, so no solution is lost. Quadrant 2: $180° - 60° = 120°$; quadrant 4: $360° - 60° = 300°$.

CONCEPT 7 · extension for the Hard band: compound angles

EQUATIONS WITH COMPOUND ANGLES (YEAR 11 ADVANCED, 5H)

In some equations, the angle is a function of x rather than simply x itself. For example,

$\tan 2x = \sqrt{3}$, where $0° \leq x \leq 360°$, or $\sin(x - 250°) = \dfrac{\sqrt{3}}{2}$, where $0° \leq x \leq 360°$.

These equations are really trigonometric equations in the *compound angles* $2x$ and $(x - 250°)$ respectively. The secret lies in solving for the *compound angle*, and in *first calculating the domain for that compound angle*.

29 EQUATIONS WITH COMPOUND ANGLES

1. Let u be the compound angle.
2. Find the restrictions on u from the given restrictions on x.
3. Solve the trigonometric equation for u.
4. Hence solve for x.

FOR THE HARD BAND ONLY

Compound angles go beyond the Year 10 course and appear only in the Hard band of this lesson. The new step is the domain: if $0° \leq x \leq 360°$, then $0° \leq 2x \leq 720°$, so $u = 2x$ goes round the circle twice and there are twice as many solutions.

TUTOR DEMO **Compound angles**

a Solve $\cos 2x = \frac{1}{2}$, where $0° \leq x \leq 360°$.

b Solve $\tan(x - 60°) = 1$, where $0° \leq x \leq 360°$.

YOUR TURN **Same method**

a Solve $\sin 2x = \dfrac{\sqrt{3}}{2}$, where $0° \leq x \leq 360°$.

b Solve $\cos(x + 45°) = -\frac{1}{2}$, where $0° \leq x \leq 360°$.

WORKED EXAMPLE 19 **From the text: keep for homework**

Solve $\tan 2x = \sqrt{3}$, where $0° \leq x \leq 360°$.

SOLUTION

Let $u = 2x.$

Then $\tan u = \sqrt{3}.$

The restriction on x is $0° \leq x \leq 360°$

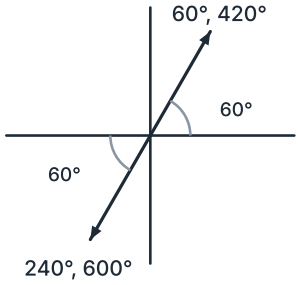
$\boxed{\times 2}$ $0° \leq 2x \leq 720°$

and replacing $2x$ by u, $0° \leq u \leq 720°.$

(The restriction on u is the key step here.)

Hence from the diagram, $u = 60°, 240°, 420°$ or $600°$.

Because $x = \frac{1}{2}u$, $x = 30°, 120°, 210°$ or $300°$.



WORKED EXAMPLE 20 **From the text: keep for homework**

Solve $\sin(x - 250°) = \frac{\sqrt{3}}{2}$, where $0° \leq x \leq 360°$.

SOLUTION

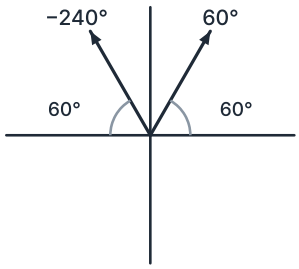
Let $u = x - 250°.$

Then $\sin u = \frac{\sqrt{3}}{2}.$

The restriction on x is $0° \leq x \leq 360°$

$\boxed{-250°}$ $-250° \leq x - 250° \leq 110°$

and replacing $x - 250°$ by u, $-250° \leq u \leq 110°.$



(Again, the restriction on u is the key step here.)

Hence from the diagram, $u = -240°$ or $60°$.

Because $x = u + 250°$, $x = 10°$ or $310°$.

STEPS THE BOOK SKIPS: WORKED EXAMPLES 19 AND 20

19 $\tan u = \sqrt{3}$ is positive, so u is in quadrant 1 or 3, related angle 60°: $u = 60°$ or $180° + 60° = 240°$. Going round again adds 360°: $420°$ and $600°$, both $\leq 720°$.

20 $\sin u$ is positive, so u is in quadrant 1 or 2, related angle 60°. In $-250° \leq u \leq 110°$, quadrant 1 gives $u = 60°$, and quadrant 2 gives $120°$, which is too big, so take $120° - 360° = -240°$. Then $x = 60° + 250° = 310°$ and $x = -240° + 250° = 10°$.

HOMEWORK · four bands, easy to past-paper level

The bands get harder: **Easy** and **Medium** practise each concept, **Hard** adds Year 11 Challenge questions and compound angles, and the **past-paper mini exam** is done last, under exam conditions. Each question names its source. Do not use a calculator when a question asks for an exact value.

EASY

1 Year 11 Advanced Exercise 5A Q10 (parts d–f omitted: they use sec, cosec and cot)

Draw the two special triangles containing the acute angles 30°, 60° and 45°. Hence write down these exact values:

a $\sin 60°$ b $\tan 30°$ c $\cos 45°$

2 Year 11 Advanced Exercise 5A Q17 (part d omitted: it uses cosec and cot)

Find without using a calculator, the value of:

a $\sin 45° \cos 45° + \sin 30°$

b $\sin 60° \cos 30° - \cos 60° \sin 30°$

c $1 + \tan^2 60°$

3 Year 11 Advanced Exercise 5G Q5 (parts b and c omitted: they use sec and cot)

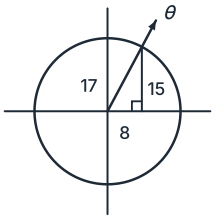
Use the complementary identities to simplify:

a $\sin(90° - \theta)$ d $\dfrac{\cos(90° - \phi)}{\sin(90° - \phi)}$

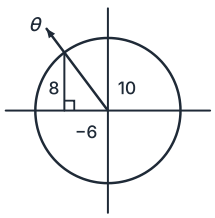
4 Year 11 Advanced Exercise 5F Q1

Write down the values of $\sin\theta$, $\cos\theta$ and $\tan\theta$ in each part.

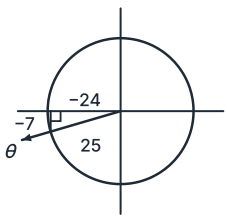
a



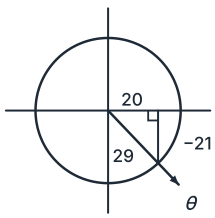
b



c



d



5 Year 11 Advanced Exercise 5F Q5

Draw similar diagrams for this question and subsequent questions.

a If $\cos\theta = \frac{1}{3}$ and θ is acute, find $\tan\theta$.

b If $\cos\theta = -\frac{4}{5}$ and θ is obtuse, find $\tan\theta$.

c If $\cos\theta = \frac{1}{2}$ and θ is reflex, find $\sin\theta$.

d Find $\cos\theta$ if $\tan\theta = -\frac{2}{3}$ and θ is reflex.

e Find $\sin\theta$, given that $\cos\theta = -\frac{40}{41}$ and $0° \leq \theta \leq 180°$.

f Find $\tan\theta$, given that $\sin\theta = \frac{1}{\sqrt{5}}$ and $-90° \leq \theta \leq 90°$.

6 Year 11 Advanced Exercise 5H Q1

Solve each equation for $0° \leq \theta \leq 360°$. (Each related acute angle is 30°, 45° or 60°.)

a $\sin\theta = \dfrac{\sqrt{3}}{2}$ b $\sin\theta = \dfrac{1}{2}$ c $\tan\theta = 1$ d $\tan\theta = \sqrt{3}$

e $\cos\theta = -\dfrac{1}{\sqrt{2}}$ f $\tan\theta = -\sqrt{3}$ g $\sin\theta = -\dfrac{1}{2}$ h $\cos\theta = -\dfrac{\sqrt{3}}{2}$

7 Year 11 Advanced Exercise 5H Q2

Solve each equation for $0° \leq \theta \leq 360°$. (The trigonometric graphs are helpful here.)

a $\sin\theta = 1$ b $\cos\theta = 1$ c $\cos\theta = 0$

d $\cos\theta = -1$ e $\tan\theta = 0$ f $\sin\theta = -1$

8 Year 11 Advanced Exercise 5H Q3

Solve each equation for $0° \leq x \leq 360°$. Use your calculator to find the related acute angle in each case, and give solutions correct to the nearest degree.

a $\cos x = \frac{3}{7}$ b $\sin x = 0.1234$ c $\tan x = 7$

d $\sin x = -\frac{2}{3}$ e $\tan x = -\frac{20}{9}$ f $\cos x = -0.77$

9 Maths Pathway, *Solving Trigonometric Equations*, Q8

Use a calculator to find every solution for x between 0° and 360° correct to two decimal places.

a) $4\sin(x) = 3$

b) $\tan(x) + 5 = 60$

c) $2\cos(x) = -1$

d) $5\tan(x) + 12 = 0$

MEDIUM

10 Year 11 Advanced Exercise 5F Q6

In this question, each part has two possible answers.

a If $\tan\alpha = \frac{1}{3}$, find $\sin\alpha$.

b If $\cos\theta = \frac{2}{\sqrt{5}}$, find $\sin\theta$.

c If $\sin\theta = \frac{3}{5}$, find $\cos\theta$.

d Find $\tan\theta$, given that $\cos\theta = -\frac{2}{3}$.

e Find $\tan\theta$, given that $\sin\theta = -\frac{12}{13}$.

f Find $\cos\theta$, given that $\tan\theta = -\frac{2}{\sqrt{3}}$.

11 Year 11 Advanced Exercise 5F Q7

a If $\cos\alpha = \frac{4}{5}$ and $\sin\alpha < 0$, find $\tan\alpha$.

b If $\tan\theta = -\frac{8}{15}$ and $\sin\theta > 0$, find $\cos\theta$.

c Find $\cos\theta$, given that $\sin\theta = \frac{1}{4}$ and $\tan\theta < 0$.

d Find $\sin\theta$, given that $\tan\theta = \frac{35}{12}$ and $\cos\theta > 0$.

e Find $\tan\theta$, given that $\sin\theta = -\frac{21}{29}$ and $\cos\theta > 0$.

f Find $\sin\theta$, given that $\cos\theta = -\frac{5}{6}$ and $\tan\theta < 0$.

12 Year 11 Advanced Exercise 5H Q4 (parts d, f and h omitted: they use cosec, sec and cot)

Solve for $0° \leq \alpha \leq 360°$. Give solutions correct to the nearest minute where necessary.

a $\sin\alpha = 0.1$ b $\cos\alpha = -0.1$ c $\tan\alpha = -1$

e $\sin\alpha = 3$ g $\sqrt{3}\tan\alpha + 1 = 0$

13 Year 11 Advanced Exercise 5H Q5 (part c omitted: it uses sec)

Solve for $-180° \leq x \leq 180°$. Give solutions correct to the nearest minute where necessary.

a $\tan x = -0.3$ b $\cos x = 0$ d $\sin x = -0.7$

14 Year 11 Advanced Exercise 5H Q6 (parts b and c omitted: they use cot and cosec)

Solve each equation for $0° \leq \theta \leq 720°$.

a $2\cos\theta - 1 = 0$

d $\tan\theta = \sqrt{2} - 1$

15 Maths Pathway, *Solving Trigonometric Equations*, Q9

Use a calculator to find the solution for x within each given range correct to two decimal places.

a) $450° < x < 720°$: $8\sin(x) = 7$

b) $360° < x < 540°$: $3\tan(x) + 4 = 40$

c) $90° < x < 270°$: $7\sin(x) = -4$

d) $-180° < x < 0°$: $11\cos(x) = 6$

16 Past paper: Girraween High School, 2023, Year 10 Mathematics Yearly Exam, Short Answer Q10 (1 mark)

Solve $\tan\theta = -\dfrac{1}{\sqrt{3}}$ for $0 \leq \theta \leq 360°$

Hint: What do you know about the value $-\dfrac{1}{\sqrt{3}}$?

17 Past paper: Hornsby Girls High School, 2021, Year 10 Mathematics Term 4 Assessment, Q16 (1 mark)

What is $\dfrac{\sin^2\theta + \cos\theta.\cos(90-\theta)}{\cos^2\theta + \sin\theta.\sin(90-\theta)}$ simplified?

A. $\sin\theta$ B. $\cos\theta$ C. $\tan\theta$ D. 1

Hint: What do you notice about the angles in the expression?

18 Past paper: Sydney Boys High School, 2024, Year 10 Stage 5 Term 4 Yearly Exam, Mathematics, Section A Q4 (2 marks)

Solve $\sin\theta = -0.5$ for $0° \leq \theta \leq 360°$.

Hint: What do you know about the value -0.5?

HARD

19 Year 11 Advanced Exercise 5A Q15 (part d omitted: it uses $\sin^2\alpha + \cos^2\alpha = 1$)

It is given that α is an acute angle and that $\tan\alpha = \dfrac{\sqrt{5}}{2}$.

a Draw a right-angled triangle (of any size), one of whose angles is α, and show this information.

b Use Pythagoras' theorem to find the length of the unknown side.

c Hence write down the exact values of $\sin\alpha$ and $\cos\alpha$.

20 Year 11 Advanced Exercise 5A Q18 (part a omitted: it uses sec)

Without using a calculator, show that:

b $2\sin 30°\cos 30° = \sin 60°$

c $\cos^2 60° - \cos^2 30° = -\frac{1}{2}$

d $\dfrac{2\tan 30°}{1-\tan^2 30°} = \tan 60°$

21 Year 11 Advanced Exercise 5F Q11

Given that $\sin\theta = \dfrac{p}{q}$, with θ obtuse and p and q both positive, find $\cos\theta$ and $\tan\theta$.

22 Year 11 Advanced Exercise 5H Q9

Solve each equation for $0° \leq \theta \leq 360°$.

a $\sin\theta = \cos\theta$

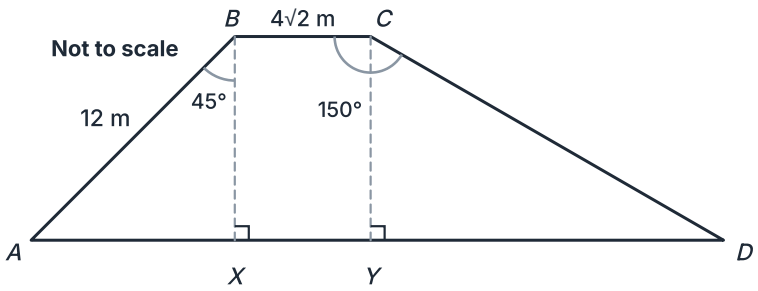
b $\sin\theta + \cos\theta = 0$

c $\sin\theta = \sqrt{3}\cos\theta$

d $\sqrt{3}\sin\theta + \cos\theta = 0$

23 Past paper: Hornsby Girls High School, 2021, Year 10 Mathematics Term 4 Assessment, Q12 (1 mark)

The diagram below shows a trapezium $ABCD$ in which BC is parallel to AD. $AB = 12\,\text{m}$, $BC = 4\sqrt{2}\,\text{m}$, $\angle ABX = 45°$ and $\angle BCD = 150°$. Perpendicular intervals are drawn from B and C to meet AD at X and Y.



What is the exact value of AD in metres?

A. $6\sqrt{2} + 6\sqrt{6}$ B. $10\sqrt{2} + 2\sqrt{6}$ C. $10\sqrt{2} + 6\sqrt{6}$ D. $16\sqrt{2} + 6\sqrt{6}$

Hint: What does the diagram give you?

24 Year 11 Advanced Exercise 5H Q7 (compound angles; part b omitted: it is Worked example 19)

Solve each equation for $0° \leq x \leq 360°$. (Let $u = 2x$.)

a $\sin 2x = \frac{1}{2}$

c $\cos 2x = -\dfrac{1}{\sqrt{2}}$

d $\sin 2x = -1$

25 Year 11 Advanced Exercise 5H Q8 (compound angles)

Solve each equation for $0° \leq \alpha \leq 360°$. (Let u be the compound angle.)

a $\tan(\alpha - 45°) = \dfrac{1}{\sqrt{3}}$

b $\sin(\alpha + 30°) = -\dfrac{\sqrt{3}}{2}$

c $\cos(\alpha + 60°) = 1$

d $\cos(\alpha - 75°) = -\dfrac{1}{\sqrt{2}}$

26 Past paper: Sydney Grammar School, 2020, Annual Examination Form IV Mathematics, Q8(c) (3 marks)

Solve $\tan(\alpha - 45°) = -\dfrac{1}{\sqrt{3}}$ for $0° \leq \alpha \leq 360°$.

Hint: What does the question give you?

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **19 marks**

Calculator allowed unless a question says otherwise · show working · hints are below each question if you get stuck

PP 1 **Girraween High School, 2023, Year 10 Mathematics Yearly Exam, Short Answer Q9** **1 mark**

Find the value of x if: $\cos(30°) = \sin(x)$, where $0 < x < 90°$.

Hint: What do you notice about the two sides of the equation?

PP 2 **North Sydney Girls High School, 2024, Year 10 Mathematics Yearly Examination, Q10** **1 mark**

Which expression is equivalent to $\dfrac{\cos(90° - \theta)}{\sin(180° - \theta)}$?

A. 1
B. −1
C. 0
D. 0.5

Hint: What do you notice about the angles in the expression?

PP 3 **Northern Beaches Christian School, 2023, Mathematics Year 10 5.3/Year 9 Accelerated Yearly Examination, Section I Q7** **1 mark**

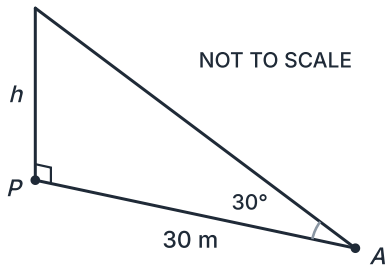
For θ between 0° and 360°, what are the values of θ for which $\sin\theta = -\frac{1}{2}$?

(A) $\theta = 210°$ and $\theta = 330°$
(B) $\theta = 30°$ and $\theta = 150°$
(C) $\theta = 30°$ and $\theta = 210°$
(D) $\theta = 150°$ and $\theta = 330°$

Hint: What do you know about the value $-\frac{1}{2}$?

PP 4 **Northern Beaches Christian School, 2022, Mathematics Year 10 5.3/Year 9 Accelerated Yearly Examination, Section I Q16** 1 mark

A point A is 30 m from point P, which is the base of a flagpole. If the top of the flagpole is at an angle of elevation of 30° from A, how tall is the flagpole?



(A) $30\sqrt{3}$ (B) $10\sqrt{3}$

(C) $\dfrac{10}{\sqrt{3}}$ (D) $15\sqrt{3}$

Hint: What does the diagram tell you?

PP 5 **North Sydney Girls High School, 2023, Year 10 Mathematics Yearly Examination, Q16(d)** 2 marks

Find the exact value of $\tan 150° + \sin 45°$.
Write your answer as a simplified fraction with rational denominator.
Show all working. Calculator answers will not be awarded credit.

Hint: What do you know about each term?

PP 6 **Sydney Grammar School, 2020, Annual Examination Form IV Mathematics, Q7(b)** 2 marks

If $\cos\theta = \dfrac{2}{\sqrt{5}}$, find the possible values of $\sin\theta$.

Hint: What does the question give you?

PP 7 **North Sydney Girls High School, 2023, Year 10 Mathematics Yearly Examination, Q17(b)** 2 marks

Solve $1 + \sqrt{3}\tan\theta = 2$ for $0° \leq \theta \leq 360°$.

Hint: What do you notice about the equation?

PP 8 **Sydney Boys High School, 2020, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics Stage 5.3, Q28** 2 marks

Solve $\sin(180° - \theta) = \sin(90° - \theta)$ for θ where $0° \leq \theta \leq 360°$.

Hint: What do you notice about the two sides of the equation?

PP 9 **Girraween High School, 2022, Year 10 Yearly Examination Mathematics, Q25(a)** 4 marks

Find all the possible values of θ, where $0° \leq \theta \leq 360°$. Give answers correct to the nearest degree.

(i) $\sin\theta = 0.416$ (ii) $\cos\theta = -0.302$

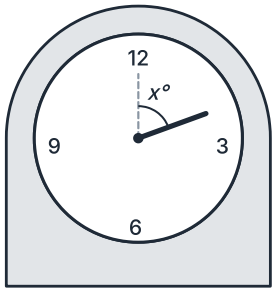
Hint: What do you know about each value?

PP 10 **Sydney Boys High School, 2024, Year 10 Stage 5 Term 4 Yearly Exam, Mathematics, Part C, Q6** 3 marks

Elsa has a grandfather clock in her house.
The height of the tip of the hour hand above the floor, in centimetres, is given by

$$h = 20\cos x^\circ + 147,$$

where x° is the angle the hour hand has rotated through since 12 o'clock.
Calculate the first 2 values of x for which the tip of the hour hand is 150 cm above the floor.



Hint: What does the question give you?

LESSON T3

The ambiguous case

When two sides and the included angle are known, the area formula gives the area; when the area is known, it gives a side or an angle, and an angle found this way has two possible values. When two sides and a *non-included* angle are known, the sine rule gives one value for the sine of the unknown angle, and there may be two triangles: this is the ambiguous case. The angle sum decides whether the obtuse answer is possible. Source: Year 11 Advanced book 5I (and two Challenge questions from 5J). The lesson ends with a chapter review of Lessons T1 to T3.

REVISION FROM TOPIC 11 (SEE THE REVISION PAGES AT THE FRONT OF THIS PACK)

In any triangle ABC, with side a opposite A, side b opposite B and side c opposite C:

- **Sine rule** (Box 30): $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$. To find a side, place the unknown side at the top left (Box 31).
- **Cosine rule** (Boxes 35 and 39): $a^2 = b^2 + c^2 - 2bc\cos A$, or $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$.
- **Area formula** (Box 32): $\text{area} = \frac{1}{2}bc\sin A$, half the product of two sides times the sine of the included angle.
- From Lesson T1: $\sin(180° - \theta) = \sin\theta$. So an acute angle and its supplement have the same sine.

CONCEPT 1 · the area formula in the SAS situation

USING THE AREA FORMULA — THE SAS CONGRUENCE SITUATION (YEAR 11 ADVANCED, 5I)

The area formula requires the SAS congruence situation in which two sides and the included angle are known.

33 USING THE AREA FORMULA

In the SAS congruence situation:

area = (half the product of two sides) × (sine of the included angle).

TUTOR DEMO **Exact area**

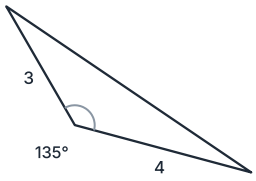
Find the exact area of a triangle with sides 5 and 8 and an included angle of 120°.

YOUR TURN **Same method**

Find the exact area of a triangle with sides 6 and 10 and an included angle of 150°.

WORKED EXAMPLE 23 **From the text: keep for homework**

Find the area of the triangle shown to the right.



SOLUTION

Using the formula, area $= \frac{1}{2} \times 3 \times 4 \times \sin 135°$.

Because $\sin 135° = \dfrac{1}{\sqrt{2}}$, (as in Example 22)

$$\text{area} = 6 \times \frac{1}{\sqrt{2}}$$

$$= 6 \times \frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \qquad \text{(rationalise the denominator)}$$

$$= 3\sqrt{2} \text{ square units.}$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 23

Example 22 is in Lesson T2. 135° is in quadrant 2, where sine is positive, and its related angle is 45°, so $\sin 135° = \sin 45° = \dfrac{1}{\sqrt{2}}$. Also $\frac{1}{2} \times 3 \times 4 = 6$, and $6 \times \dfrac{\sqrt{2}}{2} = 3\sqrt{2}$.

CONCEPT 2 · using the area formula to find a side or an angle

USING THE AREA FORMULA TO FIND A SIDE OR AN ANGLE (YEAR 11 ADVANCED, 5I)

Substituting into the area formula when the area is known may allow an unknown side or angle to be found.

When finding an angle, the formula will always give a single answer for $\sin\theta$. There will be two solutions for θ, however, one acute and one obtuse.

WHY BOTH ANGLES WORK HERE

With two sides and the included angle, any angle between 0° and 180° makes a triangle. The two answers θ and $180° - \theta$ have the same sine (Lesson T1), so they give two different triangles with the same area. The only exception is $\sin\theta = 1$, which gives the single angle 90°.

TUTOR DEMO **A side, then an angle, from the area**

a A triangle has area 40 m², and two of its sides, x m and 15 m, include an angle of 58°. Find x, correct to four significant figures.

b A triangle has area 30 cm², and two of its sides are 10 cm and 9 cm. Find the angle θ between these two sides, correct to the nearest minute.

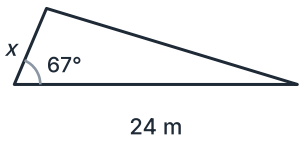
YOUR TURN **Same method**

a A triangle has area 50 m², and two of its sides, x m and 20 m, include an angle of 64°. Find x, correct to four significant figures.

b A triangle has area 21 cm², and two of its sides are 8 cm and 7 cm. Find the angle θ between these two sides, correct to the nearest minute.

WORKED EXAMPLE 24 **From the text: keep for homework**

Find x, correct to four significant figures, given that the triangle to the right has area 72 m².



SOLUTION

Substituting into the area formula,

$$72 = \tfrac{1}{2} \times 24 \times x \times \sin 67^\circ$$
$$72 = 12 \times x \times \sin 67^\circ$$
$$\boxed{\div 12 \sin 67^\circ} \quad x = \frac{6}{\sin 67^\circ}$$
$$\doteqdot 6.518 \text{ metres.}$$

WORKED EXAMPLE 25 **From the text: keep for homework**

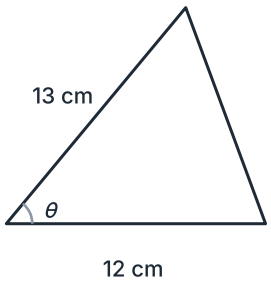
Find θ, correct to the nearest minute, given that the triangle below has area 60 cm².

SOLUTION

Substituting into the area formula,

$$60 = \tfrac{1}{2} \times 13 \times 12 \times \sin\theta$$
$$60 = 6 \times 13 \sin\theta$$
$$\boxed{\div (6 \times 13)} \quad \sin\theta = \tfrac{10}{13}.$$

Hence $\theta \doteqdot 50^\circ 17'$ or $129^\circ 43'$.



Notice that the second angle is the supplement of the first.

STEPS THE BOOK SKIPS: WORKED EXAMPLES 24 AND 25

24 $\frac{1}{2} \times 24 = 12$. Dividing both sides by $12 \sin 67°$ gives $x = \dfrac{72}{12 \sin 67°} = \dfrac{6}{\sin 67°}$, and $\sin 67° \div 0.9205$, so $x \div 6.518$.

25 $\frac{1}{2} \times 13 \times 12 = 78 = 6 \times 13$, so $\sin \theta = \dfrac{60}{78} = \dfrac{10}{13}$. The calculator gives $\sin^{-1} \frac{10}{13} \div 50.28° \div 50°17'$. The supplement is $180° - 50°17' = 129°43'$. Both are possible, because θ is the included angle.

CONCEPT 3 · the sine rule in the ambiguous ASS situation

USING THE SINE RULE TO FIND AN ANGLE — THE AMBIGUOUS ASS SITUATION (YEAR 11 ADVANCED, 5I)

The SAS congruence test requires that the angle be included between the two sides. When two sides and a non-included angle are known, the resulting triangle may not be determined up to congruence, and two triangles may be possible. This situation may be referred to as 'the ambiguous ASS situation'.

When the sine rule is applied in the ambiguous ASS situation, there is only one answer for the sine of an angle. There may be two possible solutions for the angle itself, however, one acute and one obtuse.

34 USING THE SINE RULE TO FIND AN ANGLE

In the ambiguous ASS situation, in which two sides and a non-included angle of the triangle are known,

$$\frac{\text{sine of unknown angle}}{\text{its opposite side}} = \frac{\text{sine of known angle}}{\text{its opposite side}}.$$

Always check the angle sum to see whether both answers are possible.

TUTOR DEMO **Two possible angles**

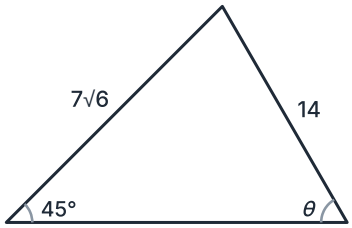
In ΔABC, $\angle B = 45°$, $AC = 10$ and $BC = 5\sqrt{6}$. Find the possible sizes of $\angle A$, and check that each is possible.

YOUR TURN **Same method**

In ΔPQR, $\angle Q = 30°$, $PR = 4$ and $QR = 4\sqrt{2}$. Find the possible sizes of $\angle P$, and check that each is possible.

WORKED EXAMPLE 26 **From the text: keep for homework**

Find the angle θ in the triangle drawn to the right.



SOLUTION

$\dfrac{\sin\theta}{7\sqrt{6}} = \dfrac{\sin 45°}{14}$ (always place the unknown at the top left)

$\sin\theta = 7 \times \sqrt{6} \times \dfrac{1}{14} \times \dfrac{1}{\sqrt{2}}$, since $\sin 45° = \dfrac{1}{\sqrt{2}}$,

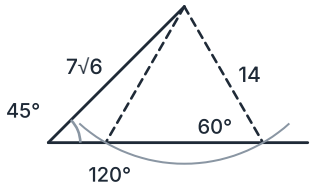
$\sin\theta = \dfrac{\sqrt{3}}{2}$

$\theta = 60°$ or $120°$.

Note: There are *two* angles whose sine is $\dfrac{\sqrt{3}}{2}$, one of them acute and the other obtuse. Moreover,

$$120° + 45° = 165°$$

leaving just 15° for the third angle in the obtuse case, so it all seems to work. Opposite is the ruler-and-compasses construction of the triangle, showing how two different triangles can be produced from the same given ASS measurements.



In many examples, however, the obtuse angle solution can be excluded using the fact that the angle sum of a triangle cannot exceed 180°.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 26

Multiply both sides by $7\sqrt{6}$: $\sin\theta = \dfrac{7\sqrt{6}\sin 45°}{14}$. Then $\dfrac{7}{14} = \dfrac{1}{2}$ and $\dfrac{\sqrt{6}}{\sqrt{2}} = \sqrt{3}$, so $\sin\theta = \dfrac{\sqrt{3}}{2}$. The related angle is 60°, and sine is positive in quadrants 1 and 2, so $\theta = 60°$ or $180° - 60° = 120°$. In the construction, the arc of radius 14 about the top vertex cuts the base line twice: once where the angle is 60° and once where it is 120°.

CONCEPT 4 · checking whether the obtuse answer is possible

THE ANGLE-SUM TEST

Suppose the sine rule gives an acute answer α for the unknown angle, and the known angle is K. The obtuse answer $180° - \alpha$ is possible only if $(180° - \alpha) + K < 180°$, that is, only if $\alpha > K$. Put another way, a larger side is always opposite a larger angle: if the known angle is opposite the *longer* of the two given sides, the unknown angle must be smaller than it, so only the acute answer is possible.

TUTOR DEMO **Only one solution**

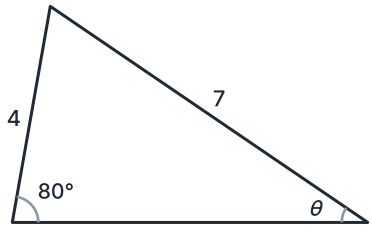
In ΔABC, $\angle C = 75°$, $AB = 8$ and $BC = 5$. Find $\angle A$, correct to the nearest minute, and show that there is only one solution.

YOUR TURN **Same method**

In ΔXYZ, $\angle Z = 70°$, $XY = 9$ and $YZ = 6$. Find $\angle X$, correct to the nearest minute, and show that there is only one solution.

WORKED EXAMPLE 27 **From the text: keep for homework**

Find the angle θ in the triangle drawn to the right, and show that there is only one solution.



SOLUTION

$\dfrac{\sin\theta}{4} = \dfrac{\sin 80°}{7}$ (always place the unknown at the top left)

$\sin\theta = \dfrac{4\sin 80°}{7}$

$\theta \doteqdot 34°15'$ or $145°45'$.

But $\theta \doteqdot 145°45'$ is impossible, because the angle sum would then exceed 180°,

so $\theta \doteqdot 34°15'$ is the only solution.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 27

$\sin\theta = \dfrac{4\sin 80°}{7} \doteqdot 0.5628$, and $\sin^{-1} 0.5628 \doteqdot 34.25° = 34°15'$. The obtuse answer is $180° - 34°15' = 145°45'$, and $145°45' + 80° = 225°45'$, which is more than 180°. Notice that the known angle 80° is opposite the longer given side 7, as in the angle-sum test above.

HOMEWORK · four bands, easy to past-paper level

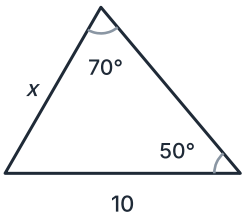
The bands get harder: **Easy** practises the sine rule and the area formula, **Medium** brings in the ambiguous case and finding angles from the area, **Hard** has the Year 11 Challenge questions and harder past-paper parts, and the **past-paper mini exam** is done last, under exam conditions. Each question names its source. A calculator is needed unless a question asks for an exact value.

EASY

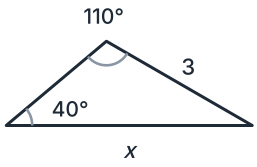
1 Year 11 Advanced Exercise 5I Q1

Find x in each triangle, correct to one decimal place.

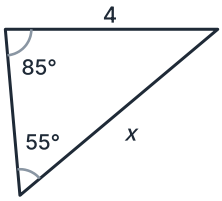
a



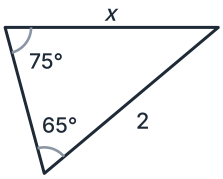
b



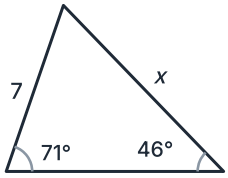
c



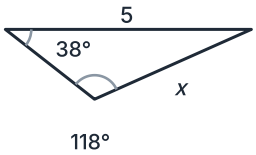
d



e



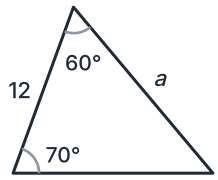
f



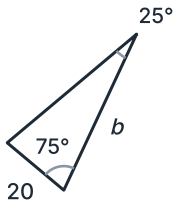
2 Year 11 Advanced Exercise 5I Q2

Find the value of the pronumeral in each triangle, correct to two decimal places.

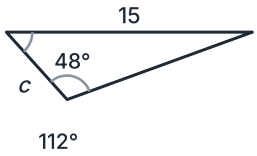
a



b



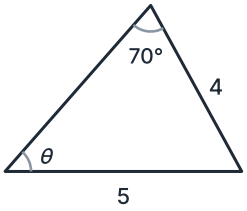
c



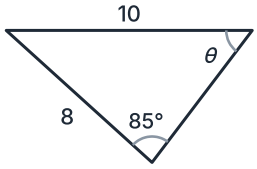
3 Year 11 Advanced Exercise 5I Q3

Find θ in each triangle, correct to the nearest degree.

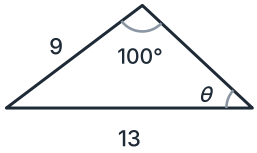
a



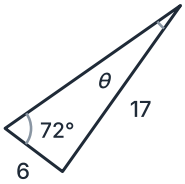
b



c



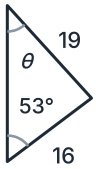
d



e



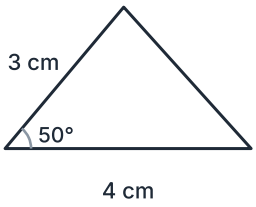
f



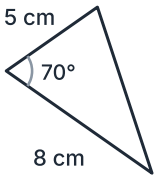
4 Year 11 Advanced Exercise 5I Q4

Find the area of each triangle, correct to the nearest square centimetre.

a



b



c



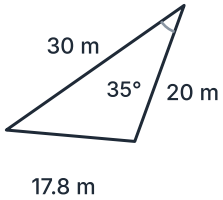
5 Past paper: Northern Beaches Christian School, 2022, Mathematics Year 10 5.3/Year 9 Accelerated Yearly Examination, Section I Q5 (1 mark)

What is the area of the triangle to the nearest square metre?

(A) 102 m² (B) 153 m²

(C) 172 m² (D) 178 m²

The diagram on the paper is marked NOT TO SCALE.



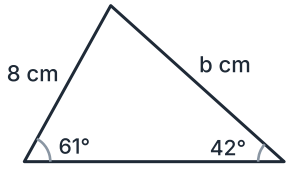
Hint: What does the diagram give you?

6 Past paper: Sydney Boys High School, 2020, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics Stage 5.3, Q20 (3 marks)

Given the diagram below, find the following, rounding your answers to 2 significant figures:

a. The value of b 1

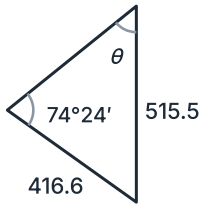
b. The area of the triangle 2



Hint: What does the diagram give you?

7 Past paper: Girraween High School, 2022, Year 10 Yearly Examination Mathematics, Q25(b) (3 marks)

Find the size of angle θ. Give answer correct to the nearest minute.



Hint: What does the diagram give you?

MEDIUM

8 Year 11 Advanced Exercise 5I Q5

a Sketch ΔABC in which $A = 43°$, $B = 101°$ and $a = 7.5$ cm.

b Find b and c, in cm correct to two decimal places.

9 Year 11 Advanced Exercise 5I Q6

a Sketch ΔXYZ in which $y = 32$ cm, $Y = 58°$ and $Z = 52°$.

b Find the perimeter of ΔXYZ, correct to the nearest centimetre.

10 Year 11 Advanced Exercise 5I Q7

Sketch ΔABC in which $a = 2.8$ cm, $b = 2.7$ cm and $A = 52°21'$.

a Find B, correct to the nearest minute.

b Hence find C, correct to the nearest minute.

c Hence find the area of ΔABC in cm², correct to two decimal places.

11 Year 11 Advanced Exercise 5I Q8

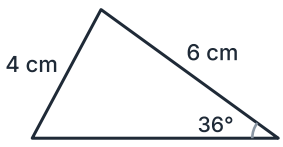
There are two triangles that have sides 9 cm and 5 cm, and in which the angle opposite the 5 cm side is 22°. Find, in each case, the size of angle opposite the 9 cm side, correct to the nearest degree.

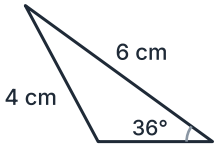




12 Year 11 Advanced Exercise 5I Q9

Two triangles are shown, with sides 6 cm and 4 cm, in which the angle opposite the 4 cm is 36°. Find, in each case, the angle opposite the 6 cm side, correct to the nearest degree.





13 Year 11 Advanced Exercise 5I Q10

Sketch ΔPQR in which $p = 7$ cm, $q = 15$ cm and $\angle P = 25°50'$.

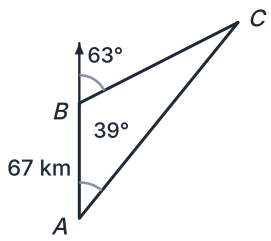
a Find the two possible sizes of $\angle Q$, correct to the nearest minute.

b For each possible size of $\angle Q$, find r in cm, correct to one decimal place.

14 Year 11 Advanced Exercise 5I Q11

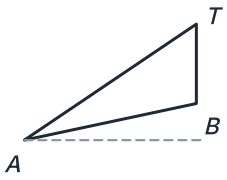
A travelling salesman drove from town A to town B, then to town C, and finally directly home to town A. Town B is 67 km north of town A, and the bearings of town C from towns A and B are 039°T and 063°T respectively.

Find how far the salesman drove, correct to the nearest kilometre.



15 Year 11 Advanced Exercise 5I Q12

Melissa is standing at A on a path that leads to the base B of a vertical flagpole. The path is inclined at 12° to the horizontal, and the angle of elevation of the top T of the flagpole from A is 34°.



a Explain why $\angle TAB = 22°$ and $\angle ABT = 102°$.

b Given that $AB = 20$ m, find the height of the flagpole, correct to the nearest metre.

16 Year 11 Advanced Exercise 5I Q13

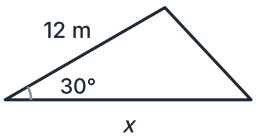
a In ΔABC, $\sin A = \frac{1}{4}$, $\sin B = \frac{2}{3}$ and $a = 12$. Find the value of b.

b In ΔPQR, $p = 25$, $q = 21$ and $\sin Q = \frac{3}{5}$. Find the value of $\sin P$.

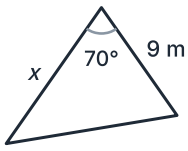
17 Year 11 Advanced Exercise 5I Q14

Substitute into the area formula to find the side length x, given that each triangle has area 48 m². Give your answers in exact form, or correct to the nearest centimetre.

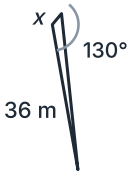
a



b



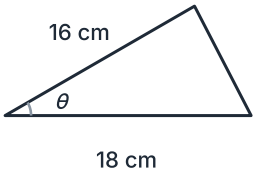
c



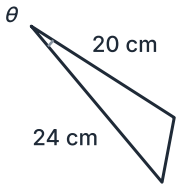
18 Year 11 Advanced Exercise 5I Q15

Substitute into the area formula to find the angle θ, given that each triangle has area 72 cm². Give answers correct to the nearest minute, where appropriate.

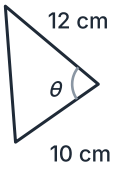
a



b



c



19 Past paper: North Sydney Boys High School, 2022, Stage 5.3 Mathematics Year 10 Assessment Task 4, Section V Q10 (4 marks)

In ΔABC, sides $c = 24$, $b = 31$ and angle $ABC = 60°$.

a) Draw ΔABC, labelling with the information given 1

b) Find angle ACB, correct to the nearest minute 3

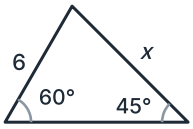
Hint: What does the question give you?

HARD

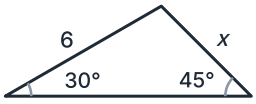
20 Year 11 Advanced Exercise 5I Q16

Find the exact value of x in each diagram.

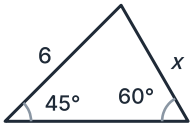
a



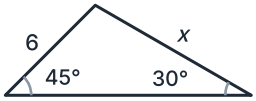
b



c



d



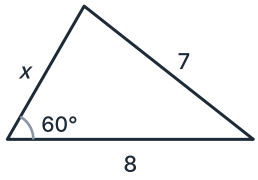
21 Year 11 Advanced Exercise 5I Q17

The diagram to the right shows an isosceles triangle in which the apex angle is 35°.
Its area is 35 cm².
Find the length of the equal sides, correct to the nearest millimetre.



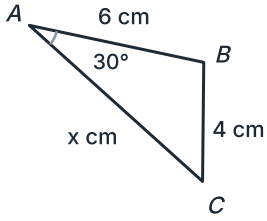
22 Year 11 Advanced Exercise 5J Q14 (Challenge)

Use the cosine rule to find the two possible values of x in the diagram to the right.



23 Year 11 Advanced Exercise 5J Q15 (Challenge)

The diagram shows ΔABC in which $\angle A = 30°$, $AB = 6$ cm and $BC = 4$ cm. Let $AC = x$ cm.

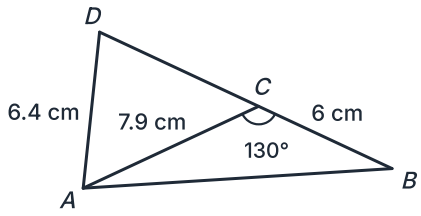


a Use the cosine rule to show that $x^2 - 6\sqrt{3}\,x + 20 = 0$.

b Use the quadratic formula to show that AC has length $3\sqrt{3} + \sqrt{7}$ cm or $3\sqrt{3} - \sqrt{7}$ cm.

c Copy the diagram and indicate on it (approximately) the other possible position of the point C.

24 Past paper: Sydney Boys High School, 2022, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part D, second Q4 (2 marks)



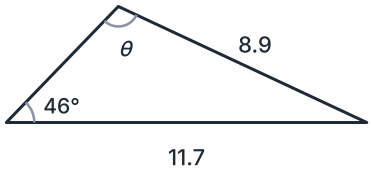
In the diagram above, BCD is a straight line.
$AC = 7.9$ cm, $AD = 6.4$ cm and $BC = 6$ cm.
Angle $ACB = 130°$.
Calculate the acute angle ADC, to the nearest degree. Geometrical reasoning not needed.

Hint: What does the diagram give you?

25 Past paper: Fort Street High School, 2022, Year 10 Common Task 2, Mathematics, Question 6(b) (3 marks)

Use the information in each triangle to find all possible values of angle θ (to the nearest degree)?

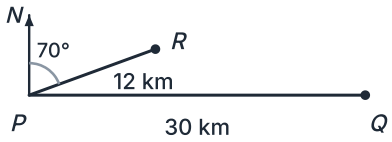
Part (a), a cosine-rule item, is not used.



Hint: What does the diagram give you?

26 Past paper: Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q22(c) (6 marks)

The diagram shows a point P which is 30 km due west of the point Q. The point R is 12km from P and has a bearing from P of 070°.



i. Find the distance of R from Q to the nearest km. 3

ii. Find the bearing of R from Q. 3

Hint: What does the question give you?

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **21 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Hornsby Girls High School, 2021, Year 10 Mathematics Term 4 Assessment, Q9** **1 mark**

In ΔDEF, $\angle D = 42°$, $d = 5$ cm and $f = 7$ cm. What is the obtuse value of $\angle F$, correct to the nearest degree?

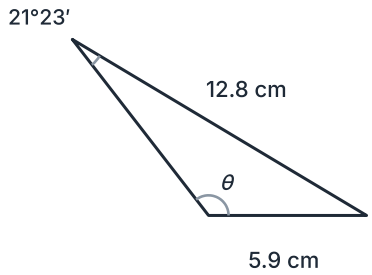
A. 28° B. 70° C. 110° D. 152°

Hint: What does the question give you?

PP 2 **The Scots College, 2023, Year 10 Assessment Task 4, Mathematics 5.3, Q19** **2 marks**

Given that θ is obtuse, find the value of θ correct to the nearest minute.

The diagram on the paper is marked Not to scale.

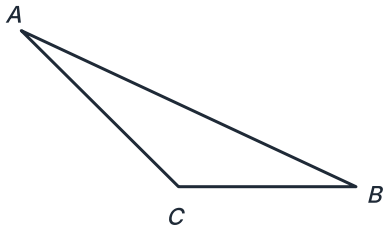


Hint: What does the diagram give you?

PP 3 **Sydney Boys High School, 2021, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part D, (b)(i)** **2 marks**

In ΔABC, $AB = 5$ cm, $AC = 3$ cm, $\angle B = 25°$, and $\angle C$ is obtuse.

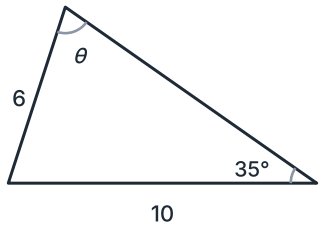
(i) Find, to the nearest degree, the size of $\angle C$.



Hint: What does the question give you?

PP 4 **Baulkham Hills High School, 2020, Year 10 Yearly Examination, Mathematics, Q14** 3 marks

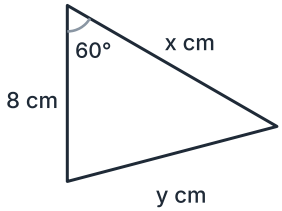
Find the value(s) of θ in the diagram. Give your answer to the nearest degree.



Hint: What does the diagram give you?

PP 5 **Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q22(b)** 6 marks

The area of the triangle is $22\sqrt{3}$ cm².



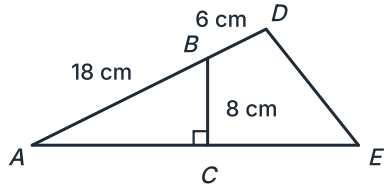
i. Find the value of x. 3

ii. Hence find the exact value of y. 3

Hint: What does the question give you?

PP 6 **Sydney Boys High School, 2024, Year 10 Stage 5 Term 4 Yearly Exam, Mathematics, Part B, Q4** 3 marks

In the diagram below, $AB = 18$ cm, $BD = 6$ cm, $BC = 8$ cm, and the area of triangle $ADE = 160$ cm².



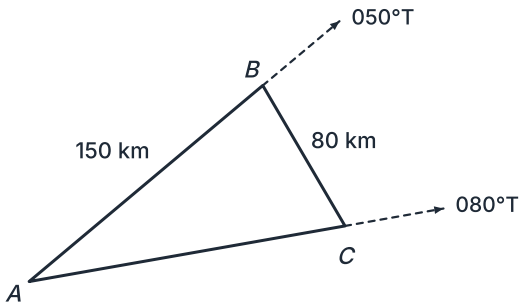
Calculate the length of AE, to 2 decimal places.

Hint: What does the diagram give you?

PP 7 **North Sydney Girls High School, 2023, Year 10 Mathematics Yearly Examination, Q20(a)** 4 marks

Pocahontas headed from A to B, 150 km away on a bearing of 050°T.
She then turned towards C, which is 80 km from B and on a bearing of 080°T from A.

NOT TO SCALE



A is further from C than from B.
Through what angle did Pocahontas turn at B?

Hint: What does the question give you?

CHAPTER REVIEW

Chapter review: Lessons T1 to T3

Mixed questions from the whole pack: angles of any size, exact values and equations, the ambiguous case of the sine rule, and the cosine and area rules. Each question is used nowhere else in this pack. Allow about 20 minutes for the 14 marks.

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1 Sydney Boys High School, 2024, Year 10 Stage 5 Term 4 Yearly Exam, Mathematics, Section A Q5 (1 mark)

What is the angle that the line $y = -x$ makes with the positive direction of the x-axis?

Hint: What does the question give you?

2 North Sydney Girls High School, 2022, Year 10 Mathematics Yearly Examination, Q18(a) (2 marks)

Find the exact value of $\cos 45° + \tan 120°$. You do not need to simplify your answer.

Hint: What do you know about each term?

3 Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q22(a) (2 marks)

Solve this trigonometric equation for $0° \leq \theta \leq 360°$. $\tan\theta = -1$

Hint: What do you notice about the equation?

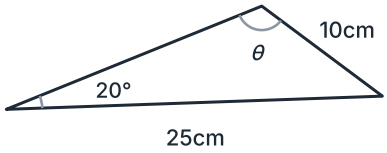
4 Sydney Boys High School, 2022, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part D, first Q4 (2 marks)

Solve $\cos x = -\cos 225°$ for $0° \leq x \leq 360°$.

Hint: What do you notice about the equation?

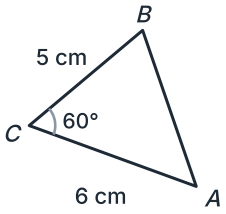
5 Fort Street High School, 2021, Year 10 Common Task 2, Mathematics, Question 6 (3 marks)

Find the value(s) of angle θ to the nearest degree.



Hint: What does the diagram give you?

6 Sydney Grammar School, 2020, Annual Examination Form IV Mathematics, Q3(b) (4 marks)



In $\triangle ABC$ above, $\angle ACB = 60°$, $BC = 5$ cm and $AC = 6$ cm.

(i) Use the cosine rule to calculate the exact length of side AB. 2

(ii) Calculate the exact area of $\triangle ABC$. 2

Hint: What does the diagram give you?